AOS-CX 10.06 IP Routing Guide

6100, 6200 Switch Series

Hewlett Packard Enterprise

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Notices

Acknowledgments

Chapter 1
About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
example-text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([ ]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: • *<example-text>* • <example-text> • *example-text* • *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
|	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Table Continued

Convention	Usage
[ ]	Brackets. Indicates that the enclosed item or items are optional.
… or . . .	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

`switch>`

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(`*`CONTEXT-NAME`*`)#`

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

`switch(config-vlan-100)#`

When referring to this context, this document uses the syntax:

`switch(config-vlan-<VLAN-ID>)#`

Where *`<VLAN-ID>`* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

`member/slot/port`

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on the switch.

On the 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 in slot 1 on member 1.

Overview

VRF is a technology that allows multiple instances of a routing table to co-exist within the same router. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other. Network functionality is improved because network paths can be segmented without requiring multiple routers.

VRF support

The 6100 and 6200 Switch Series support predefined VRFs:

- **`default`**: All other interfaces are automatically assigned to the VRF `default` (6100 and 6200 Switch Series).
- **`mgmt`**: Assigned to the management port and is used to isolate management traffic (6200 Switch Series only).

NOTE: User-defined VRFs are not supported on the 6100 and 6200 Switch Series.

Overview

NOTE: Loopback interfaces are not supported on the Aruba 6100 Switch Series.

A loopback interface is a virtual interface supporting IPv4/IPv6 address configuration. The loopback interface can be considered stable because once created, it remains up. The loopback interface can then be configured with an address to use as a reference or identifier independent of the physical interfaces.

- **Device identification**: As long as the router is operational, the state of the loopback interface is always up. Even if only one link to the router is active, the loopback interface can be reached. This functionality makes it possible to identify an active device in the network using the IP address configured on the loopback interface.
- **Routing**: Since the loopback interface is always active, a routing session (such as a BGP session) can continue on an alternate path even if the outbound interface fails. In OSPF, a loopback interface address is advertised as an interface route into the network. This functionality increases reliability by allowing traffic to take alternate paths if there is a link failure. In OSPF and BGP, the router ID can be set to the loopback address to avoid reassignment of the router ID when physical interfaces are added or removed.
- **Device management**: Loopback interface is always reachable and can be used for sending and receiving management information such as logs and SNMP traps without interruption.

Loopback commands

`interface loopback`

Syntax

```
interface loopback <INSTANCE>

no interface loopback <INSTANCE>
```

Description

Creates a loopback interface and enters loopback configuration mode.

The `no` form of this command deletes a loopback interface.

Command context

`config`

Parameters

<INSTANCE>

Selects the loopback interface ID. Range: 0 to 4

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

ip address

Syntax

`ip address <IPV4-ADDR/MASK> [secondary]`

`no ip address <IPV4-ADDR/MASK> [secondary]`

Description

Sets the IPv4 address for a loopback interface.

The `no` form of this command reverses the set of the IPv4 address for a loopback interface.

Command context

`config`

Parameters

<IPV4-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

secondary

Indicates that the IPv4 address is a secondary address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 16.93.50.2/24
switch(config-loopback-if)# ip address 20.1.1.1/24 secondary
```

ipv6 address

Syntax

`ipv6 address <IPV6-ADDR/MASK>`

Description

Sets the IPv6 address for a loopback interface.

Command context

config

Parameters

<IPV6-ADDR>

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ipv6 address fd00:5708::f02d:4df6/64
```

show interface loopback

Syntax

show interface loopback [brief | instance *<ID>*]

Description

This command displays the configuration and status of loopback interfaces.

Command context

Operator (>) or Manager (#)

Parameters

brief

Displays brief information about all configured loopback interfaces.

instance *<ID>*

Displays the configuration and status of a loopback interface ID. Range: 1-255

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show interface loopback

 Interface loopback1 is up
 IPv4 address 192.168.1.1/24

 Interface loopback2 is up
 IPv4 address 182.168.1.1/24
```

```
switch# show interface loopback brief

---------------------------------------------------------------------
Loopback      IP Address                                     Status
Interface
---------------------------------------------------------------------

loopback1   10.1.1.1/24                                         up
loopback1   1111:2222:3333:4444::6666/128                       up
```

```
switch# show interface loopback 1
 Interface loopback1 is up
 IPv4 address 192.168.1.1/24
```

Overview

Static routes are manually configured. If the network topology is simple enough, you can use static routes for the network's routing requirements. The proper configuration and usage of static routes can improve network performance and ensure bandwidth for important network applications.

The disadvantage of using static routes is that they cannot adapt to network topology changes. If a fault or topological change occurs in the network, the relevant routes will be unreachable and a network administrator must modify the static routes manually.

Default route

Without a default route, a packet that does not match any routing entries is discarded and an ICMP destination-unreachable packet is sent to the source. A default route is used to forward packets that do not match any routing entry.

The network administrator can configure a default route with the destination as 0.0.0.0 and the mask as 0. The router forwards any packet whose destination address fails to match any entry in the routing table to the next hop of the default static route.

Recursive static routes

Recursive static routes are not supported.

A recursive static route is a route for which the next hop is learned from another routing look up (for example, dynamic protocol or from another another static route). For example, the following commands create an unsupported recursive static route:

- `ip route 99.0.0.0/24 30.0.0.2` - Main static route with gateway 30.0.0.2.
- `ip route 30.0.0.0/24 20.0.0.2` - Static route to reach the nexthop of the previously configured route.

Configuration concepts

Before configuring a static route, you must understand the following concepts:

- **Destination address and mask**: In the `ip route` command, an IPv4 address is in dotted-decimal format. A mask can be in the form of mask length - the number of consecutive 1s in the mask.
- **Output interface and next hop address**: When configuring a static route, specify the output interface or next hop address. The next hop address cannot be a local interface IP address or the route configuration will not take effect.
- **Other attributes**: You can configure different priorities and administrative distance for different static routes to make route management policies more flexible. For example, specifying the same priority for different routes to the same destination enables load sharing, but specifying different priorities for these routes enables route backup.

Configuration example procedure

Configuration prerequisites

Before configuring a static route, complete the following tasks:

- Configure the physical parameters for related interfaces.
- Configure the link-layer attributes for related interfaces.
- Configure the IP addresses for related interfaces.

NOTE: 6100 Switch Series: The number of IPv4 and IPv6 static routes that can be configured is limited to 512 combined.

6200 Switch Series: The number of IPv4 and IPv6 static routes that can be configured is limited to 1024 combined. 16K routes are not supported.

6100 and 6200 Series Switches do not support IPv6 addresses with prefixes greater than 64. Prefixes between 65 and 127 will be software-forwarded.

Basic static route configuration example

The IP addresses and masks of the switches and hosts are displayed here. Static routes are required for interconnection between any two hosts.

Procedure

1. Configure IP addresses for interfaces (details not shown).
2. Configure static routes.
   a. Configure a default route on Switch A.
   ```
   6300# config
   6300(config)#
   <SwitchA> config
   [SwitchA] ip route 0.0.0.0/0 1.1.4.2
   ```
   b. Configure two static routes on Switch B.
   ```
   <SwitchB> config
   [SwitchB] ip route 1.1.2.0/24 1.1.4.1
   [SwitchB] ip route 1.1.3.0/24 1.1.5.6
   ```
   c. Configure a default route on Switch C.
   ```
   6300# config
   6300(config)#
   [SwitchC] ip route 0.0.0.0/0 1.1.5.5
   ```
3. Configure the hosts. The default gateways for hosts A, B, and C are `1.1.2.3`, `1.1.6.1`, and `1.1.3.1`. The configuration procedure is not shown.
4. Display the configuration.
   a. Display the IP routing table of Switch A.
   b. Display the IP routing table of Switch B.

c. Use the `ping` command on Host B to check the reachability of Host A, assuming Windows XP runs on the two hosts.

```
C:\Documents and Settings\Administrator>ping 1.1.2.2
Pinging 1.1.2.2 with 32 bytes of data:
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
15
Ping statistics for 1.1.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

d. Use the `traceroute` command on Host B to check the reachability of Host A.

```
8320# traceroute
traceroute  Trace the IPv4 route to a device on the network
traceroute6 Trace the IPv6 route to a device on the network
8320# traceroute
A.B.C.D Enter IPv4 address of the device to traceroute
WORD  Enter host name oo fthe device to traceroute
8320# traceroute
```

Static routing commands

ip route

Syntax

ip route *<DEST-IPV4-ADDR>/<NETMASK>* [*<NEXT-HOP-IP-ADDR>*|*<INTERFACE>*|reject|blackhole]

no ip route *<DEST-IPV4-ADDR>/<NETMASK>* [*<NEXT-HOP-IP-ADDR>*|*<INTERFACE>*|reject|blackhole]

Description

Adds an IPv4 static route on the default VRF.

The `no` form of this command deletes a IPv4 static route.

Command context

`config`

Parameters

<DEST-IPV4-ADDR>/<NETMASK>

Specifies the IPv4 route destination.

<NEXT-HOP-IP-ADDR>

Specifies the next hop address for reaching the destination in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

<INTERFACE>

Specifies the next hop as an outgoing interface.

blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip route 10.0.0.0/24 blackhole
switch(config)# ip route 10.0.1.0/24 reject
switch(config)# ip route 10.0.2.0/24 20.0.0.2
switch(config)# ip route 10.0.3.0/24 1/1/1
```

ip route distance

Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>

no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

Description

Configures the administrative distance for the IPv4 static route.

The `no` form of this command deletes the static route.

Command context

`config`

Parameters

<DEST-IPV4-ADDR>>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32.

<NEXT-HOP-IP-ADDR>

Specifies the next hop IPv4 address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

<INTERFACE>

Specifies the next hop as an outgoing interface.

distance *<VALUE>*

Specifies the administrative distance to associate with this static route. Default: 1. Range: 1-255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Not supported on the 6200 Switch Series.

```
switch(config)# ip route 10.0.2.0/24 20.0.0.2 distance 4
switch(config)# ip route 10.0.3.0/24 1/1/1 distance 6
```

```
switch(config)# no ip route 10.0.3.0/24 1/1/1 distance 6
```

ip route tag

Syntax

```
ip route <A.B.C.D/M> [A.B.C.D | INTERFACE] tag <1-4294967295>

no ip route <A.B.C.D/M> [A.B.C.D | INTERFACE] tag <1-4294967295>
```

Description

Configures tag for IPv4 static route.

The `no` form of this command deletes tag for IPv4 static route.

Command context

`config`

Parameters

tag

Specifies and assigns tag for the route.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# ip route 10.1.1.5/32 1/1/1 tag 20
```

```
switch(config)# no ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# no route 10.1.1.5/32 1/1/1 tag 20
```

ipv6 route

Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>|reject|blackhole]

no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>|reject|blackhole]
```

Description

Adds an IPv6 static route.

The `no` form of this command deletes an IPv6 static route on the default VRF.

Command context

`config`

Parameters

<DEST-IPV6-ADDR>

Specifies the route destination in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

<NEXT-HOP-IP-ADDR>

Specifies the next hop in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<INTERFACE>

Specifies the next hop as an outgoing interface.

`blackhole`

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

`reject`

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Access platforms for IPv6 routes above /64 are not supported.
- IPv6 addresses with prefixes 65-127 will not be configured in the ASIC route table.

Examples

```
switch(config)# ipv6 route 120::/124 blackhole
switch(config)# ipv6 route 121::/124 reject
switch(config)# ipv6 route 122::/124 1/1/1
switch(config)# ipv6 route 123::/124 120::1
```

```
switch(config)# no ipv6 route 122::/124 1/1/1
```

ipv6 route distance

Syntax

`ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>`

`no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>`

Description

Configures the administrative distance for the IPv6 static route

The `no` form of this command deletes the static route.

Command context

`config`

Parameters

<DEST-IPV6-ADDR>

Specifies the route destination address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

<NEXT-HOP-IP-ADDR>

Specifies the next hop in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<INTERFACE>

Specifies the next hop as an outgoing interface.

distance *<VALUE>*

Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ipv6 route 122::/124 1/1/1 distance 5
switch(config)# ipv6 route 123::/124 120::1 distance 6
```

```
switch(config)# no ipv6 route 123::/124 120::1 distance 6
```

ipv6 route tag

Syntax

`ipv6 route <X:X::X:X/M> [X:X::X:X | INTERFACE] tag <1-4294967295>`

`no ipV6 route <X:X::X:X/M> [X:X::X:X | INTERFACE] tag <1-4294967295>`

Description

Configures tag for IPv6 static route.

Command context

`config`

Parameters

tag

Specifies and assigns tag for the route.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Not supported on the 6200 Switch Series.

```
switch(config)# ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# ipv6 route 3002::1/128 1000::2 tag 20
```

```
switch(config)# no ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# no ipv6 route 3002::1/128 1000::2 tag 20
```

show ip route

Syntax

```
show ip route [<A.B.C.D> | <A.B.C.D/M> | all-vrfs | bgp | connected | local | ospf | static | summary |
     vrf<vrf-name>]
```

Description

Displays IPv4 route tables.

Command context

Operator (>) or Manager (#)

Parameters

<A.B.C.D>

Display longest prefix match.

<A.B.C.D/M>

Display exact route match.

all-vrfs

Display information for all VRFs.

bgp

(6200 Switch Series only) Display bgp routes only.

connected

Display connected routes only.

local

Display local routes only.

ospf

(6200 Switch Series only) Display ospf routes only.

static

Display static routes only.

summary

Display the aggregate count of routes per routing protocol.

vrf <vrf-name>

Specify a VRF by VRF name (if no *<VRF-NAME>* is specified, the default VRF is implied.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing IPv4 route tables:

```
switch# show ip route

Displaying ipv4 routes selected for forwarding

'[x/y]' denotes [distance/metric]

10.0.0.0/24, vrf default
        via  vlan2,  [0/0],  connected
10.0.0.1/32, vrf default
        via  vlan2,  [0/0],  local
10.100.11.0/24, vrf default
        via  vlan1,  [0/0],  connected
10.100.11.82/32, vrf default
        via  vlan1,  [0/0],  local
20.0.0.0/24, vrf default
        via  10.0.0.2,  [1/0],  static
20.0.1.0/24, vrf default
        via  10.0.0.2,  [1/0],  static
20.0.2.0/24, vrf default
        via  vlan1,  [1/0],  static
20.0.4.0/24, vrf default
        blackhole route,  [1/0],  static
20.0.5.0/24, vrf default
        reject route,  [1/0],  static
```

Showing IPv4 route tables for the test VRF:

```
switch# show ip route vrf test

Displaying ipv4 routes selected for forwarding

'[x/y]' denotes [distance/metric]

30.0.0.0/24,  1 (null) next-hops
        via  30.0.0.2,  [0/0],  connected
90.0.0.0/24,  1 unicast next-hops
        via  30.0.0.1,  [1/0],  static
90.0.1.0/24,  1 unicast next-hops
        via  1/1/2,  [1/0],  static
90.0.3.0/24, blackhole, 1, [1/0], static
```

show ipv6 route

Syntax

```
show ipv6 route [<X.X.X.X> | <X.X.X.X/M> | all-vrfs | bgp | connected | local | ospf | static | summary |
    vrf <vrf-name>]
```

Description

Displays IPv6 route tables.

Command context

Operator (>) or Manager (#)

Parameters

<X.X.X.X>

Display exact route match.

<X.X.X.X/M>

Display exact route match.

all-vrfs

Display information for all VRFs.

bgp

(6200 Switch Series only) Display bgp routes only.

connected

Display connected routes only.

local

Display local routes only.

ospf

(6200 Switch Series only) Display ospf routes only.

static

Display static routes only.

summary

Display the aggregate count of routes per routing protocol.

vrf *<vrf-name>*

Specify a VRF by VRF name (if no *<VRF-NAME>* is specified, the default VRF is implied.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing IPv6 route tables:

```
switch# show ipv6 route

Displaying ipv6 routes selected for forwarding

'[x/y]' denotes [distance/metric]

1000::/64, vrf default
        via  vlan2,  [0/0],  connected
1000::1/128, vrf default
        via  vlan2,  [0/0],  local
2000::/64, vrf default
```

```
        via  vlan2,  [1/0],  static
2001::/64, vrf default
        via  1000::2,  [1/0],  static
3000:2301::/64, vrf default
        blackhole route,  [1/0],  static
4000:2301::/64, vrf default
        reject route,  [1/0],  static
```

Overview

NOTE: OSPFv2 is not supported on the Aruba 6100 Switch Series.

Open Shortest Path First version 2 (OSPFv2) is a routing protocol described in RFC2328. It is a Link State-based IGP (Interior Gateway Protocol) routing protocol applied to routers grouped into OSPF areas identified by the routing configuration on each routing switch. It is widely used in medium to large-sized enterprise networks.

The characteristics of OSPFv2 are:

- Provides a loop-free topology using SPF algorithm.
- Supports load balancing with equal cost routes for the same destination.
- OSPFv2 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Provides authentication of routing messages.
- Provides fast convergence with triggered, incremental updates via LSAs.

Some OSPFv2 configuration is done in the global configuration context, others in the router ospf context, or in the interface configuration context, or in the vlink context. OSPFv2 can be configured on L3 ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv2 mandates the associated interface to be a routed interface.

Supported features

- OSPFv2 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- SPF throttling
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.

- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)

How OSPFv2 protocol works

OSPF protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

OSPFv2 concepts

OSPFv2 Link-state advertisement (LSA) types

OSPFv2 sends routing information in Link-State Advertisements (LSAs). The switches support the following types of LSAs.

- **Router LSA—Type-1 LSA.** Describes the states of the router interfaces to an area, and is flooded throughout a single area only.
- **Network LSA—Type-2 LSA.** Describes the list of routers connected to the network, and is flooded throughout a single area only.
- **Summary LSA—Type-3 LSA.** Describes the route to networks in another OSPF area of the same Autonomous System (AS). Propagated through backbone area to other areas.
- **ASBR summary LSA—Type-4 LSA.** Describes the route to an ASBR in an OSPF normal or backbone area of the same AS. Propagated through backbone area to other areas.
- **AS external LSA—Type-5 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.
- **NSSA LSA—Type-7 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in NSSA. ABR converts type-7 LSAs to type-5 LSAs for injection into the backbone area.

OSPFv2 router types

Internal routers

Internal OSPFv2 routers belong to only one area. Internal routers flood type-1 LSAs to all routers in the same area and maintain identical LSDBs.

Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding summary LSAs between its border areas. You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IP addresses. This aggregate address is advertised instead of all the individual addresses it represents.

Autonomous system boundary router (ASBR)

Autonomous system boundary routers run multiple interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPF through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

Designated routers (DRs)

In an OSPF network having two or more routers, one router is elected to serve as the DR and another router to act as the Backup Designated Router (BDR). All other routers in the area forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This action minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPF network with no DR and no BDR, the neighboring router with the highest priority is elected the DR, and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPF network are declaring themselves DRs, both priority and router ID are used to select the DR and BDRs.

Priority is configurable using the `ip ospf priority` command at the interface level. If two neighbors share the same priority, the router with the highest router ID is elected as the DR. The router with the next highest router ID is elected as the BDR.

OSPFv2 area types

OSPFv2 is built upon a hierarchy of network areas. All areas for a given OSPF domain reside in the same AS. An AS is defined as a number of contiguous networks that share the same interior gateway routing protocol.

An AS can be divided into multiple areas. Each area represents a collection of contiguous networks and hosts, and the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which types 1 and 2 LSAs are broadcast, which limits the amount of LSA flooding that occurs within the AS and also helps to control the size of the LSDBs maintained in OSPF routers. An area is represented in OSPF by either an IP address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and stub.

Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured virtual link. The backbone is a transit area that carries the type-3 summary LSAs, type-5 AS external link LSAs and routed traffic between non-backbone areas, as well as the type-1 and type-2 LSAs and routed traffic internal to the area. ASBRs are allowed in backbone areas.

Normal area

Normal area connects to the backbone area through one or more ABRs (physically or through a virtual link) and supports type-3 summary LSAs and type-5 external link LSAs to and from the backbone area. ASBRs are allowed in normal areas.

Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow external (type 5) LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise summary routes from other areas.
- Use the default summary (type-3) route to advertise both of the following:
  - Summary routes to other areas in the AS
  - External routes to other ASs

You can configure the stub area ABR to do the following:

- Suppress advertising from some or all area summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing type-3 summary LSAs and the default external route from the backbone area with the default summary route (0.0.0.0/0.)

Virtual links are not allowed for stub areas.

Not-so-stubby (NSSA) area

NSSA area connects to the backbone area through one or more ABRs. NSSAs are intended for use where an ASBR exists in an area where you want to control the following:

- Advertising the ASBR external route paths to the backbone area
- Allowing LSAs from the backbone area to advertise in the NSSA:
  - Summary routes (type-3 LSAs) from other areas
  - External routes (type-5 LSAs) from other areas as a default external route (type-7 LSAs)

Virtual links are not allowed for NSSAs.

OSPFv2 configuration task list

Tasks at a glance

- **Enabling OSPF on the routing switch**
- **Assigning the routing switch to an OSPF area**
- **Setting OSPF network for the area**
- **Creating an OSPF virtual link for an area**
- **Configuring external route redistribution and control**
- **Configuring ranges on an ABR to reduce advertising to the backbone**

- **Influencing route choice by changing the administrative distance**
- **Configuring graceful restart of OSPF routing**
- **Configuring OSPF interface settings**
- **Configuring OSPF interface authentication**
- **Configuring all OSPF interfaces as passive**
- **Viewing OSPFv2 information**
- **Clearing OSPF statistics on a switch**

Configuring OSPF on the routing switch

Create the OSPF instance and enter the OSPF router configuration context. From this, you can proceed with other OSPF configuration tasks.

Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPF instance and enter the OSPF router configuration context.
- To configure a router ID, create OSPF network areas, or adjust other global OSPF configuration items, you must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

1. Create the OSPF instance and enter the OSPF router configuration context using the following command. For command details, see **`router ospf`**.

   `router ospf <PROCESS-ID>`

   For example, the following command creates OSPF instance 1.

   ```
   switch(config)# router ospf 1
   switch(config-ospf-1)#
   ```
2. Configure a global router ID using the following command. For command details, see **`router-id`**.

   `router-id <ROUTER-ADDRESS>`

   For example, the following command sets the router ID to 1.1.1.1.

   ```
   switch(config-ospf-1)# router-id 1.1.1.1
   ```
3. Optionally, if the OSPF process was disabled (it is enabled by default), enable the OSPF process using the following command.

   `enable`

   For command details, see **`enable`**. (Refer to **`disable`** for disabling the OSPF process).

Assigning the routing switch to an OSPF area

Create a Normal, Stub, or Not So Stubby (NSSA) area.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

1. Create an OSPF area for the routing switch using one of the following commands:
   - Create a normal area using the following command: `area <area-id>`. For command details, see **area**.
   - Create a stub area using the following command: `area <area-id> stub`. For command details, see .
   - Create a Not-So-Stubby-Area using the command: `area <area-id> nssa`. For command details, see **area nssa**.

   For example, the following command creates a normal area with an area identifier of 10.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

   ```
   switch(config-ospf-1)# area 10.1.1.1
   ```

Setting OSPF network for the area

After you define an OSPF area, you can assign one or more networks to it. The OSPF protocol will run on the interface with the configured IPv4 address. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

1. Set an OSPF network for the area using the following command. For command details, see **ip ospf area** .

   ```
   ip ospf <PROCESS-ID> area <AREA-ID>
   ```

   For example, use the following command to assign interface 1/1/1 to OSPF area 1. The area can alternatively be entered as an IPv4 address.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ip ospf 1 area 1
   ```

2. Optionally, you can disable OSPF on the interface using the following command. For command details, see **ip ospf shutdown**.

   ```
   ip ospf shutdown
   ```

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ip ospf shutdown
   ```

Configuring external route redistribution and control

Configuring route redistribution for OSPF establishes the routing switch as an ASBR for importing and translating different protocol routes into OSPF. When you configure redistribution for OSPF, you can specify that static, connected, or BGP routes external to the OSPF domain are imported as OSPF routes.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

1. Enable route redistribution using the command **redistribute** .

   `redistribute {bgp | connected | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]`

   For example, the following command sets redistribution of connected routes as OSPF routes:

   ```
   switch(config-ospf-1)# redistribute connected
   ```

2. Optionally, modify the default metric for redistribution using the following command. For command details, see **default-metric**.

   `default-metric <METRIC-VALUE>`

   For example, the following command sets the default metric for redistribution to 37.

   ```
   switch(config-ospf-1)# default-metric 37
   ```

3. Optionally, set the cost of default-summary LSAs using the following command. For command details, see **area default-metric**.

   `area <AREA-ID> default-metric <COST>`

   For example, the following command sets the cost of default summary LSAs to 2.

   ```
   switch (config-ospf-1)# area 1 default-metric 2
   ```

4. Optionally, set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. For command details, see **max-metric router-lsa**.

   `max-metric router-lsa [on-startup [<ADVERT-TIME>]]`

   For example, the following command sets advertise max-metric router-lsa on startup.

   ```
   switch(config-ospf-1)# max-metric router-lsa on-startup 3000
   ```

5. Set the maximum number of ECMP routes that OSPF can support using the following command. For command details, see .

   `maximum-paths <MAX-VALUE>`

   For example, the following command sets the maximum number of ECMP routes to 8.

   ```
   switch(config-ospf-1)# maximum-paths 8
   ```

Influencing route choice by changing the administrative distance

The administrative distance value can be left in its default configuration setting (110) unless a change is needed to improve OSPF performance for a specific network configuration.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure the administrative distance using the following command. For command details, see **distance** command.

```
distance <distance>
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospf-1)# distance 100
```

Configuring graceful restart of OSPF routing

OSPF routing can be gracefully restarted on the switch without losing packets that are in transit. OSPF neighbors are informed that the router is completing a graceful restart, which allows for maintenance on the switch without interrupting traffic in the network. There is no effect on the saved switch configuration.

Prerequisites

NOTE: Graceful restart is only applicable to the 6200-VSF switches.

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Enable graceful restart of OSPF routing using the following command. For command details, see **graceful-restart**.

```
graceful-restart {restart-interval <seconds> | helper}
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospf-1)# graceful-restart restart-interval 50
```

Configuring OSPF interface settings

You can optionally adjust the following OSPF interface settings.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

- Set the interface cost using the following command. For command details, see **ip ospf cost**.

  ```
  ip ospf cost <interface-cost>
  ```

  For example, the following command sets the cost associated with interface 1/1/1 to 100.

  ```
  switch(config)# interface 1/1/1
  switch(config-if)# ip ospf cost 100
  ```

- Set the time interval between OSPF hello packets for the OSPF interface using the following command. For command details, see **ip ospf hello-interval**.

  ```
  ip ospf hello-interval <seconds>
  ```

- Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface using the following command. For command details, see **ip ospf dead-interval**.

  `ip ospf dead-interval <seconds>`

- Set the time between retransmitting lost link state advertisements for the OSPF interface using the following command. For command details, see **ip ospf retransmit-interval**.

  `ip ospf retransmit-interval <seconds>`

- Sets the transit delay in link state transmission for the OSPF interface using the following command. For command details, see **ip ospf transit-delay**.

  `ip ospf transit-delay <seconds>`

- Set the OSPF network type for the interface. For command details, see **ip ospf network**.

  `ip ospf network {broadcast|point-to-point}`

- Set the OSPF priority on the interface using the following command. For command details, see **ip ospf priority**.

  `ip ospf priority <number-value>`

- Set the OSPF interface as OSPF passive interface. The interface participates in OSPF but does not send or receive packets on that interface. For command details, see **ip ospf passive**.

  `ip ospf passive`

Configuring OSPF interface authentication

Configure authentication on the interface. Only one method of authentication can be active on an interface at a time. If one method is already configured on an interface, configuring an alternative method on the same interface automatically overwrites the first method used.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

1. Set the authentication type that will be used for authentication with the neighbor router using the following command. For command details, see .

   `ip ospf authentication {message-digest | simple-text | null | keychain}`

   For example, the following command sets the authentication type to simple-text.

   ```
   switch(config-if)# ip ospf authentication simple-text
   ```

2. For simple-text authentication, set the password using the following command. For command details, see **ip ospf authentication-key**.

   `ip ospf authentication-key <PASSWORD>`

   For example:

```
switch(config-if)# ip ospf authentication-key secure
```

3. For MD5 authentication, set the password using the following command. For command details, see **ip ospf message-digest-key md5**.

   ```
   ip ospf message-digest-key {Key ID} md5 {ciphertext | plaintext} <KEY>
   ```

   For example:

   ```
   switch(config-if)# ip ospf message-digest-key 1 md5 ciphertext 100
   ```

Configuring all OSPF interfaces as passive

OSPF sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPF to be passive.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Configure all OSPF interfaces as passive using the following command. For command details, see **passive-interface default**.

```
passive-interface default
```

```
switch(config-ospf-1)# passive-interface default
```

Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure SPF throttling timers using the following command. For command details, see **timers throttle spf**

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-
time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time
3000 max-wait-time 9000
```

Viewing OSPFv2 information

Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

Procedure

1. To view OSPFv2 information, use the following commands. For command details and examples, click the links.
    - To view general OSPF, area, state and configuration information use: **`show ip ospf`**.
    - To view OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR) use: **`show ip ospf border-routers`**.
    - To view OSPF link state database summary for different OSPF LSAs (Link State Advertisement) use: **`show ip ospf lsdb`**.

      Many different parameters are available with this command to display information for a particular LSA.
    - To view information about OSPF-enabled interfaces use: **`show ip ospf interface`**.
    - To view information about OSPF neighbors use: **`show ip ospf neighbors`**.
    - To view OSPF routing table information use: **`show ip ospf routes`**.
    - To view OSPF statistics use: **`show ip ospf statistics`**.
    - To view OSPF statistics for the OSPF-enabled interfaces use **`show ip ospf statistics interface`**.
    - To view the current state and parameters of the OSPF virtual-links use **show ip ospf virtual-links** .

Clearing OSPF statistics on a switch

Procedure

Clear OSPF event statics using the following command. For command details, see **`clear ip ospf statistics`**.

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs |
vrf <VRF-NAME>]
```

For example, the following command clears OSPF event statistics from interface 1/1/1.

```
switch(config-router)# clear ip ospf statistics interface 1/1/1
```

An example of the OSPFv2 information in the show running-config command

When OSPFv2 is configured, the output of the `show running-config` command includes OSPFv2 information.

Example

```
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.XXXX
!
lldp enable
timezone set utc
mgmt
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf vrf_default
    area 0.0.0.0
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf hello-interval 22
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf passive
    ip ospf dead-interval 88
    ip ospf transit-delay 44
    ip ospf priority 88
    ip ospf cost 8
    ip ospf network point-to-point
    ip ospf authentication simple-text
    ip ospf authentication-key kkk
    ip ospf shutdown
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip ospf 1 area 0.0.0.0
```

OSPFv2 commands

area

Syntax

`area <AREA-ID>`

`no area <AREA-ID>`

Description

Creates a normal area, with `<AREA-ID>` set if not present. If the area is already present and it is not a normal area, then this command changes the area type to normal.

The `no` form of this command deletes the area with the `<AREA-ID>` specified. Area can be of any type (nssa, nssa no-summary, stub, stub no-summary, and default normal area).

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<AREA-ID>

Specifies the area ID in one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a normal area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1
Switch(config-ospf-1)# show running-config current-context router ospf 1
    router-id 1.1.1.1
    area 0.0.0.1
```

Deleting an area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1
```

area default-metric

Syntax

`area <AERA-ID> default-metric <COST>`

`no area <AREA-ID> default-metric`

Description

Sets the cost of the default route announced to NSSA or stub areas.

The `no` form of this command resets the cost of the default route announced to NSSA or stub areas, to the default value of 1.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<AREA-ID>

Specifies area ID in one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

default-metric *<COST>*

Sets the cost of default-summary LSAs announced to NSSA or stub areas, to the specified value. Default cost: 1. Range: 0 to 16777215.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting cost for default LSA summary:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 default-metric 2
```

Setting cost for default LSA summary to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 default-metric
```

area nssa

Syntax

`area <AREA-ID> nssa [no-summary]`

`no area <AREA-ID> nssa [no-summary]`

Description

Creates the NSSA area (Not So Stubby Area) with *<AREA-ID>* if not present. If area is present and not NSSA area, this command changes the area type to NSSA area. If `no-summary` is used, area type will be NSSA No-Summary.

The `no` form of this command unsets the area type as NSSA. That is, the configured area will be changed to default normal area. The `no area <AREA-ID> nssa no-summary` command enables sending inter-area routes into NSSA, but will not unset the area as NSSA.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<AREA-ID>

Specifies the area ID in one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

`nssa [no-summary]`

Specifies Not So Stubby Area (NSSA) area type. If area is present and not NSSA area, parameter changes the area type to NSSA area. If `no-summary` is specified, area type will be NSSA No-Summary, which means do not inject inter-area routes into NSSA.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an NSSA area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 nssa
switch(config-ospf-1)# area 1 nssa no-summary
```

Unsetting the area as NSSA

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 nssa
switch(config-ospf-1)# no area 1 nssa no-summary
```

clear ip ospf neighbors

Syntax

```
clear ip ospf [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Resets the neighbor and clears the OSPF neighbor information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies the OSPFv2 process ID to clear the statistics for the particular OSPFv2 process. Range: 1 to 63.

<NEIGHBOR>

Specifies the router ID of a neighbor.

<INTERFACE-NAME>

Specifies the OSPFv2 statistics to clear for the specified interface.

all-vrfs

Select to clear the OSPFv2 statistics for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Clearing the OSPFv2 neighbor information:

```
switch# clear ip ospf 1 neighbor 3.3.3.3
switch# clear ip ospf 1 neighbor interface 1/1/1
```

clear ip ospf statistics

Syntax

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-NAME>]
```

Description

Clear the OSPF event statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

OSPF process ID. Clear the statistics for the particular OSPF process. Range: 1 to 63.

<INTERFACE-NAME>

Clear the OSPF statistics for the specified interface.

all-vrfs

Optionally select to clear the OSPF statistics for all VRFs.

vrf *<VRF-NAME>*

Optionally select to clear the OSPF statistics for a particular VRF. If the VRF is not specified, information for the default VRF is cleared.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing the OSPF event statistics:

```
switch# clear ip ospf statistics
switch# clear ip ospf statistics interface 1/1/1
```

default-information originate

Syntax

```
default-information originate

no default-information originate
```

Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table.

The `no` form of this command disables advertisement of the default route.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

default-information originate always

Syntax

```
default-information originate always

no default-information originate always
```

Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

default-metric

Syntax

`default-metric <METRIC-VALUE>`

`no default-metric`

Description

Sets the default metric for redistributed routes in the OSPF.

The `no` form of this command sets the default metric to be used for redistributed routes into OSPF to the default of 25.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<METRIC-VALUE>

Specifies the default metric value to use for redistributed routes. Default: 25. Range: 0-1677214.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-metric 37
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-metric
```

disable

Syntax

`disable`

Description

Disables the OSPF process.

IMPORTANT: This command does not remove the OSPF configurations.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# disable
```

distance

Syntax

```
distance <DISTANCE>
```

```
no distance
```

Description

Defines an administrative distance for OSPF. Administrative distance is used as a criteria to select the best route when multiple routes are present from different routing protocols.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

Command context

```
config-ospf-<PROCESS-ID>
```

Parameters

<DISTANCE>

Specifies OSPF administrative distance. Range: 1-255. Default: 110.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF administrative distance:

```
switch(config)# router ospf 1
switch(config-ospf-1)# distance 100
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no distance
```

enable

Syntax

```
enable
```

Description

Enables the OSPF process, if disabled. By default the OSPF process is enabled.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# enable
```

graceful-restart

Syntax

`graceful-restart {restart-interval <SECONDS> | helper}`

`no graceful-restart {restart-interval | helper}`

Description

Configures graceful restart parameters for OSPF.

The `no` form of this command sets the restart interval to the default interval of 120 seconds or disables the helper mode, depending on the parameter specified.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

Choose one of the following parameters:

`restart-interval <SECONDS>`

Specifies the time another router should wait for this router to gracefully restart and selects the maximum interval (in seconds) that another router should wait. Default: 120 seconds. Range: 5-1800.

`helper`

Specifies that the router will participate in the graceful restart of a neighbor router.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling OSPF nonstop forwarding:

```
switch(config)# router ospf 1
switch(config-ospf-1)# graceful-restart restart-interval 50
switch(config-ospf-1)# graceful-restart helper
```

Setting restart-interval to default, disable helper mode:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no graceful-restart restart-interval
switch(config-ospf-1)# no graceful-restart helper
```

ip ospf area

Syntax

```
ip ospf <PROCESS-ID> area <AREA-ID>

no ip ospf <PROCESS-ID> area <AREA-ID>
```

Description

Runs the OSPF protocol on the interface with the configured IPv4 address for the area specified. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The `no` form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol.

Command context

`config-if`

`config-if-vlan`

Parameters

<PROCESS-ID>

Specifies the OSPF process Id. Range: 1 to 63.

<AREA-ID>

Specifies the OSPF area ID in one of the following formats.

- Area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- Area identifier in decimal format. Range: 0 to 4294967295.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf 1 area 1
switch(config-if-vlan)# ip ospf 1 area 0.0.0.1
```

Disabling OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf 1 area 1
switch(config-if-vlan)# no ip ospf 1 area 0.0.0.1
```

ip ospf authentication

Syntax

```
ip ospf authentication {message-digest | simple-text | null | keychain}
```

```
no ip ospf authentication
```

Description

Sets the authentication type that will be used for authentication with the neighbor router.

The `no` form of this command deletes the authentication type used for a particular authentication with the neighbor router and sets to null authentication.

Command context

config-if

config-if-vlan

Parameters

Choose one of the following authentication types.

`message-digest`

Sets authentication type as message-digest.

`simple-text`

Sets authentication type as simple-text.

`null`

Sets authentication type as null.

`keychain`

Sets the authentication type to use the key chain.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF authentication type on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication simple-text
```

Deleting OSPF authentication type on the interface and sets it to null:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication
```

ip ospf authentication-key

Syntax

```
ip ospf authentication-key {ciphertext | plaintext} <PASSWORD>
```

```
no ip ospf authentication-key
```

Description

Sets the authentication password used for simple-text authentication. If the password is given in ciphertext it will be decrypted and applied to the protocol.

The `no` form of this command deletes the authentication password used for simple-text authentication.

Command context

`config-if`

`config-if-vlan`

Parameters

`ciphertext`

Specifies providing the password in encrypted format.

`plaintext`

Specifies providing the password in simple-text format.

`<PASSWORD>`

Specifies the password for simple-text authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF simple-text authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key ciphertext AQBapdAVzEZb205RmH33+4cEctMayhvmuOVeYmXs8Y7ANu2eBgAAABO5xecopg==
switch(config-if-vlan)# ip ospf authentication-key plaintext secure
```

Deleting OSPF simple-text authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication-key
```

ip ospf cost

Syntax

`ip ospf cost <INTERFACE-COST>`

`no ip ospf cost`

Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The `no` form of this command sets the cost (metric) associated with a particular interface to the default cost 1.

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERFACE-COST>

Specifies the interface cost value. Range: 1 to 65535. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF interface cost

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf cost 100
```

Setting the OSPF interface cost to default

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf cost
```

ip ospf dead-interval

Syntax

`ip ospf dead-interval <INTERVAL>`

`no ip ospf dead-interval`

Description

Sets the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface.

The `no` form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPF interface. The default value is 40 seconds (generally 4 times the hello packet interval).

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERVAL>

Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf dead-interval 30
```

Setting OSPF dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf dead-interval
```

ip ospf hello-interval

Syntax

`ip ospf hello-interval <INTERVAL>`

`no ip ospf hello-interval`

Description

Sets the time interval between OSPF hello packets for the OSPF interface.

The `no` form of this command sets the time interval OSPF hello packets to the default of 10 seconds for the OSPF interface.

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERVAL>

Specifies the time interval for the hello interval, in seconds. Range: 1 to 65535. Default: 10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf hello-interval 30
```

Setting OSPF hello interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf hello-interval
```

ip ospf keychain

Syntax

`ip ospf keychain <KEYCHAIN-NAME>`

`no ip ospf keychain`

Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The `no` form of this command deletes the key chain used for md5 authentication.

Command context

`config-if`

Parameters

<KEYCHAIN-NAME>

Name of key chain to be used for md5 authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf keychain ospf_keys
```

Deleting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf keychain
```

ip ospf message-digest-key md5

Syntax

`ip ospf authentication message-digest`

`ip ospf message-digest-key <KEY-ID> md5 {ciphertext | plaintext} <KEY>`

`no ip ospf message-digest-key <KEY-ID>`

Description

Sets the message digest password used for message digest authentication. If the md5 key is given in ciphertext, it will be decrypted and applied to the protocol.

The `no` form of this command deletes the message digest password used for message digest authentication.

Command context

`config-if`

`config-if-vlan`

Parameters

`ciphertext`

Specifies providing the md5 key in encrypted format.

`plaintext`

Specifies providing the md5 key in simple-text format.

<KEY-ID>

Specifies the password for md5 (message-digest) authentication. Range: 1 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF message-digest authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 ciphertext AQBapdAVzEZb205RmH33+4cEctMayhvmuOVeYmXs8Y7ANu2eBgAAABO5xecopg==
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 plaintext secure
```

Deleting OSPF message-digest authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf message-digest-key 1
```

ip ospf network

Syntax

`ip ospf network {broadcast | point-to-point}`

`no ip ospf network`

Description

Configures the OSPF network type for the interface.

The `no` form of this command sets the network type for the interface to the system default which is broadcast network.

Command context

`config-if`

`config-if-vlan`

Parameters

Choose one of the following parameters as the interface network type.

broadcast

Specifies the OSPF network type as a broadcast multi-access network.

point-to-point

Specifies the OSPF network type as a point-to-point network.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf network broadcast
switch(config-if-vlan)# ip ospf network point-to-point
```

Disabling OSPF network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf network
```

`ip ospf passive`

Syntax

`ip ospf passive`

`no ip ospf passive`

Description

Configures the interface as an OSPF passive interface. With this setting, the interface participates in OSPF but does not send or receive packets on that interface.

The `no` form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

Command context

`config-if`

`config-if-vlan`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the interface as OSPF passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf passive
```

Setting the interface as OSPF active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf passive
```

`ip ospf priority`

Syntax

`ip ospf priority <PRIORITY-VALUE>`

`no ip ospf priority`

Description

Sets the OSPF priority for the interface. The larger the numeric value of the priority, the higher the chances for it to become the designated router. Setting a priority of zero makes the router ineligible to become a designated router or back up designated router.

The `no` form of this command sets the OSPF priority for the interface to the default of 1.

Command context

`config-if`

`config-if-vlan`

Parameters

<PRIORITY-VALUE>

Specifies the OSPF priority value. Range: 0 to 255. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF priority for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf priority 50
```

Disabling OSPF priority for the interface to default:

```
switch(config)# interface vlan1
switch(config-if-vlan)# no ip ospf priority
```

ip ospf retransmit-interval

Syntax

`ip ospf retransmit-interval <INTERVAL>`

`no ip ospf retransmit-interval`

Description

Sets the time between retransmitting lost link state advertisements for the OSPF interface.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPF interface.

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERVAL>

Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf retransmit-interval 30
```

Setting OSPF retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf retransmit-interval
```

ip ospf shutdown

Syntax

`ip ospf shutdown`

`no ip ospf shutdown`

Description

Disables OSPF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command `no ip ospf area`.

The `no` form of this command re-enables OSPF on the interface

Command context

`config-if`

`config-if-vlan`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf shutdown
```

Re-enabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf shutdown
```

ip ospf transit-delay

Syntax

`ip ospf transit-delay <DELAY>`

`no ip ospf transit-delay`

Description

Sets the time delay in link state transmission for the OSPF interface.

The `no` form of this command sets the delay in link state transmission to the default of 1 second for the OSPF interface.

Command context

`config-if`

`config-if-vlan`

Parameters

<DELAY>

Specifies the transit delay in seconds. Range: 1 to 3600. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF transit delay on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf transit-delay 30
```

Setting OSPF transit delay to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf transit-delay
```

keychain

Syntax

`keychain <KEYCHAIN-NAME>`

`no keychain`

Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The `no` form of this command deletes the key chain used for md5 authentication.

Command context

`config-router-vlink`

Parameters

<KEYCHAIN-NAME>

Name of key chain to be used for md5 authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF virtual link key chain authentication:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# keychain ospf_keys
```

Deleting OSPF virtual link key chain authentication:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link  100.0.1.1
switch(config-router-vlink)# no keychain
```

max-metric router-lsa

Syntax

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]

no max-metric router-lsa [on-startup]
```

Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup for the time mentioned in seconds or for a default value of 600 seconds.

The `no` form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

`on-startup <ADVERT-TIME>`

Specifies the time in seconds to advertise self as stub-router on startup. If no time is specified, the default time of 600 seconds is used. Range: 5 to 86400. Default: 600.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospf 1
switch(config-ospf-1)# max-metric router-lsa
switch(config-ospf-1)# max-metric router-lsa on-startup
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no max-metric router-lsa
switch(config-ospf-1)# no max-metric router-lsa on-startup
```

passive-interface default

Syntax

```
passive-interface default

no passive-interface
```

Description

Configures all OSPF interfaces as passive.

The `no` form of this command sets all OSPF interfaces as active.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF-enabled interfaces as passive:

```
switch(config)# router ospf 1
switch(config-ospf-1)# passive-interface default
```

Setting OSPF-enabled interfaces as active:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no passive-interface
```

redistribute

Syntax

`redistribute {bgp | connected | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]`

`no redistribute {bgp | connected | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]`

Description

Redistributes routes originating from other protocols, or from another OSPF process, to the current OSPF process.

The `no` form of this command disables redistribution of routes to the current OSPF process.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

bgp

Specifies redistributing BGP (Border Gateway Protocol) routes.

connected

Specifies redistributing connected (directly attached subnet or host).

static

Specifies redistributing static routes.

rip

Specifies redistributing RIP routes.

ospf *<PROCESS-ID>*

Specifies redistributing routes from the specified OSPFv2 process ID.

route-map *<ROUTE-MAP-NAME>*

Specifies the name of a route map. To create a route map, use the `route-map` command.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Redistributing routes to OSPF:

```
switch(config)# router ospf 1
switch(config-ospf-1)# redistribute connected
switch(config-ospf-1)# redistribute static
switch(config-ospf-1)# redistribute static route-map static_networks
switch(config-ospf-1)# redistribute rip
switch(config-ospf-1)# redistribute ospf 2
```

Disabling redistributing routes to OSPF:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no redistribute connected
switch(config-ospf-1)# no redistribute static
switch(config-ospf-1)# no redistribute rip
switch(config-ospf-1)# no redistribute ospf 2
```

reference-bandwidth

Syntax

`reference-bandwidth <BANDWIDTH>`

`no reference-bandwidth`

Description

Sets the reference bandwidth for OSPFv2. If the OSPFv2 interface cost is not explicitly set. then the cost of all the OSPFv2 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the link speed value is taken as 1 Gbps (if the OSPFv2 interface cost is not explicitly set).

The `no` form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<BANDWIDTH>

Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the reference bandwidth:

```
switch(config)# router ospf 1
switch(config-ospf-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no reference-bandwidth
```

rfc1583-compatibility

Syntax

`rfc1583-compatibility`

`no rfc1583-compatibility`

Description

Enables OSPF compatibility with RFC1583 (backward compatibility). If RFC1583 compatibility is enabled, then the route cost calculation follows a different method.

The `no` form of this command disables OSPF compatibility with RFC1583 (backward compatibility). By default the RFC1583 compatibility is disabled.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# rfc1583-compatibility
```

Disabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no rfc1583-compatibility
```

router ospf

Syntax

`router ospf <PROCESS-ID> [vrf <VRF-NAME>]`

`no router ospf <PROCESS-ID> [vrf <VRF-NAME>]`

Description

Creates an OSPF process (if not created already) on a VRF, and switches to the OSPF router instance context. Up to eight OSPF processes are supported per VRF.

The `no` form of this command removes the OSPF instance.

Command context

`config`

Parameters

<PROCESS-ID>

Specifies an OSPF process ID. Range: 1 to 63.

vrf *<VRF-NAME>*

Specifies a VRF name for the OSPF process. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# router ospf 1
switch(config-ospf-1)#
```

```
switch(config)# router ospf 1 vrf vrf_red
```

```
switch(config)# no router ospf 1
```

router-id

Syntax

`router-id <ROUTER-ADDR>`

`no router-id`

Description

Sets an ID for the router in an IPv4 address format.

The `no` form of this command unconfigures the router-id for the instance and sets the router-id to the default as follows: the router-id is selected dynamically as equal to the highest loopback address on the router, or the highest active interface if there are no loopback addresses. If no IP address is configured on any interfaces on the router, OSPF will not form an adjacency.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<ROUTER-ADDR>

Specifies the Router ID in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting router-id in the OSPF context:

```
switch(config)# router ospf 1
switch(config-ospf-1)# router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no router-id
```

show ip ospf

Syntax

```
show ip ospf [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays general OSPF, area, state, and configuration information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63.

all-vrfs

Optionally select to display general OSPF information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing general OSPF configurations:

```
switch# show ip ospf 1
Routing Process 1 with ID : 1.1.1.1 VRF default
------------------------------------------------

OSPFv2 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Summary address:
  prefix 20.1.1.0/24 advertise tag 10
Number of areas is 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is 1, 1 normal, 0 stub, 0 NSSA
BFD is disabled
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
```

```
  SPF calculation has run 7 times
  Area ranges:
  Number of LSAs: 3, checksum sum 75093
Area (0.0.0.1) (Inactive)
  Interfaces in this Area: 0 Active Interfaces: 0
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 2 times
  Area ranges:
    ip-prefix 20.1.1.1/24, inter-area, advertise
  Number of LSAs: 0, checksum sum 0
```

show ip ospf border-routers

Syntax

```
show ip ospf [<PROCESS-ID>]
     border-routers [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays the OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display the OSPF routing table entries for ABR and ASBR for the particular OSPF process. Range: 1 to 63.

all-vrfs

Select to display OSPF border router information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPF border routers information:

```
switch# show ip ospf border-routers
OSPF Process ID 1 VRF default Internal Routing Table
-----------------------------------------------------

Codes: i - Intra-area route, I - Inter-area route

i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 15 via
    10.1.1.2, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 15 via
    11.1.1.12, Interface 1/1/2
```

show ip ospf interface

Syntax

```
show ip ospf [<PROCESS-ID>] interface [<interface-name>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Displays general OSPF, area, state, and configuration information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63.

***<interface-name>*]**

Specify the name of an OSPF interface.

brief

Display a brief overview of OSPF configuration information.

all-vrfs

Select to display general OSPF information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing general OSPF configuration settings for the default VRF:

```
switch# show ip ospf 1
Routing Process 1 with ID : 1.1.1.1 VRF default
------------------------------------------------

OSPFv2 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Summary address:
  prefix 20.1.1.0/24 advertise tag 10
Number of areas is 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is 1, 1 normal, 0 stub, 0 NSSA
```

```
BFD is disabled
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 7 times
  Area ranges:
  Number of LSAs: 3, checksum sum 75093
Area (0.0.0.1) (Inactive)
  Interfaces in this Area: 0 Active Interfaces: 0
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 2 times
  Area ranges:
    ip-prefix 20.1.1.1/24, inter-area, advertise
  Number of LSAs: 0, checksum sum 0
```

Showing OSPF configuration settings for all interfaces:

```
switch# show ip ospf interface
Interface 1/1/1 is up, line protocol is up
-------------------------------------------

IP address 10.1.1.1/24, Process ID 1 VRF default, area 0.0.0.0
    State Backup-dr, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit delay 1 sec, Router priority 1
    Designated Router IP: 10.1.1.2
    Backup Designated Router IP: 10.1.1.1
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    No authentication
    Number of Link LSAs: 0, checksum sum 0
    BFD is disabled

Interface 1/1/2 is up, line protocol is up
-------------------------------------------

IP address 11.1.1.1/24, Process ID 1 VRF default, area 0.0.0.1
    State Dr, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit delay 1 sec, Router priority 1
    Designated Router IP: 11.1.1.1
    No backup designated router on this network
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    No authentication
    Number of Link LSAs: 0, checksum sum 0
    BFD is disabled
```

show ip ospf lsdb

Syntax

```
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     asbr-summary [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<router-id> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
```

```
     external [lsid <LINK-STATE-ID>] [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     router [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     network [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     summary [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     nssa-external [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     database-summary
```

Description

Shows the OSPF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display the OSPF link state database information for the specified OSPF process. Range: 1 to 63.

all-vrfs

Select to display OSPF link state database information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Optionally select one of the following parameters to filter the link state database information.

- `asbr-summary`. Displays ASBR summary link states (LSA type 4).
- `external`. Displays external link states (LSA type 5). The external parameter does not take the area <AREA-ID> parameter.
- `router`. Displays router LSAs (LSA type 1).
- `network`. Displays network LSAs (LSA type 2).
- `summary`. Displays network-summary link states (LSA type 3).
- `nssa-external`. Displays NSSA external link states (LSA type 7).

database-summary

Select to display the count of each type of LSA and each area in the database.

area *<AREA-ID>*

Select to display information filtered for the specified area in one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Value: 0 to 4294967295.

lsid *<LINK-STATE-ID>*

Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).

adv-router {*<ROUTER-ID>* | self}

Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify `self` to show self-originated link states.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPF link state database (LSDB) general information:

```
switch# show ip ospf lsdb
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
============================================================

Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------

LSID            ADV Router      Age     Seq#        Checksum    Link Count
---------------------------------------------------------------------------
40.40.40.40     40.40.40.40     930     0x80000004  0x2ea1      3
50.50.50.50     50.50.50.50     935     0x80000002  0x8b52      1
60.60.60.60     60.60.60.60     943     0x800003c5  0x9854      2

Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------

LSID            ADV Router      Age     Seq#        Checksum
--------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001  0x7179
192.0.2.1       50.50.50.50     935     0x80000001  0x516a

Inter-area Summary Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------

LSID            ADV Router      Age     Seq#        Checksum
--------------------------------------------------------------
209.165.201.1   40.40.40.40     929     0x80000001  0x2498
209.165.201.1   50.50.50.50     928     0x80000001  0x5b2f
209.165.201.1   60.60.60.60     1265    0x800003c3  0xf49b
192.0.2.0       40.40.40.40     943     0x80000001  0x53f3
192.0.2.0       50.50.50.50     935     0x80000001  0x26f8
192.0.2.0       60.60.60.60     930     0x80000001  0x7b51
```

Showing ASBR summary link states:

```
switch# show ip ospf lsdb asbr-summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

ASBR Summary Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------

LSID            ADV Router      Age     Seq#            Checksum
-------------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001      0x7179
192.0.2.1       50.50.50.50     935     0x80000001      0x516a
```

Showing external link states:

```
switch# show ip ospf lsdb external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

AS External Link State Advertisements
--------------------------------------

LSID            ADV Router      Age     Seq#            Checksum
-------------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001      0x7179
192.0.2.1       50.50.50.50     935     0x80000001      0x516a
```

Showing database summary:

```
switch# show ip ospf lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
==========================================================

Area 0.0.0.0 database summary
------------------------------

LSA Type             Count
---------------------------
Router               2
Network              1
Inter-area Summary   1
ASBR Summary         0
NSSA External        0
Subtotal             4

Process 1 database summary
---------------------------

LSA Type             Count
---------------------------
Router               2
Network              1
Inter-area Summary   1
ASBR Summary         0
NSSA External        0
AS External          0
Total                4
```

Showing router LSAs:

```
switch# show ip ospf lsdb router
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

Router Link State Advertisements (Area 0.0.0.0)
```

```
-----------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum Link Count
-------------------------------------------------------------------------
1.1.1.2         1.1.1.2         15          0x80000004 0xf526    1
2.2.2.1         2.2.2.1         14          0x80000005 0x6c5e    2
2.2.2.2         2.2.2.2         104         0x80000004 0xf51a    1
```

Showing network LSAs:

```
switch# show ip ospf lsdb network
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
1.1.1.2         1.1.1.2         141         0x80000001 0xc55e
2.2.2.2         2.2.2.2         230         0x80000001 0xa179
```

Showing network-summary link states:

```
sswitch# show ip ospf lsdb summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

Inter-area Summary Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
1.1.1.0         2.2.2.1         133         0x80000002 0xa089

Inter-area Summary Link State Advertisements (Area 0.0.0.1)
-----------------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
2.2.2.0         2.2.2.1         133         0x80000002 0x7caa
```

Showing NSSA external link states:

```
switch(config-ospf-1)# show ip ospf lsdb nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

NSSA External Link State Advertisements (Area 0.0.0.1)
------------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
8.8.8.0         1.1.1.2         162         0x80000003 0xc7b2
```

show ip ospf neighbors

Syntax

```
show ip ospf [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
     [interface <INTERFACE-NAME>] [detail | summary]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays information about OSPF neighbors.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 63.

neighbors *<NEIGHBOR-ID>*

Select to display information about a particular neighbor, specified in IPv4 format (A.B.C.D).

interface *<INTERFACE-NAME>*

Select to display neighbor information only for the specified interface.

detail

Select to display detailed information for all the neighbors.

summary

Select to display summary information for the neighbors.

all-vrfs

Select to display neighbor information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPF neighbors information for the default VRF:

```
switch# show ip ospf neighbors
OSPF Process ID 1 VRF default
==============================

Total Number of Neighbors: 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
2.2.2.2          1         FULL/DR           10.1.1.2           1/1/1

```

Showing OSPF neighbors information for VRF red:

```
switch# show ip ospf neighbors vrf red
OSPF Process ID 1 VRF red
==========================

Total Number of Neighbors: 1
```

```
Neighbor ID      Priority  State              Nbr Address       Interface
-------------------------------------------------------------------------
1.1.1.1          1         FULL/BDR           10.1.1.1           1/1/1
```

Showing OSPF neighbors information for a specific neighbor:

```
switch# show ip ospf neighbors 2.2.2.2
Neighbor 2.2.2.2, interface address 10.1.1.2
--------------------------------------------
Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
Options is 0x42
Dead timer due in 00:00:32
Time since last state change 00h:19m:17s
```

Showing OSPF neighbors information for a specific neighbor and interface:

```
switch# show ip ospf neighbors 2.2.2.2 interface 1/1/1
Neighbor 2.2.2.2, interface address 10.1.1.2
--------------------------------------------
Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
DR is 10.1.1.2, BDR is 10.1.1.1
Options is 0x42
Dead timer due in 00:00:38
Retransmission queue length 0
Time since last state change 00h:22m:21s
```

Showing detail information for OSPF neighbors:

```
switch# show ip ospf neighbors detail
Neighbor 2.2.2.2, interface address 10.1.1.2
--------------------------------------------
Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
DR is 10.1.1.2, BDR is 10.1.1.1
Options is 0x42
Dead timer due in 00:00:38
Retransmission queue length 0
Time since last state change 00h:22m:21s
```

Showing summary information for OSPF neighbors in default VRF:

```
switch# show ip ospf neighbors summary
OSPF Process ID 1 VRF default, Neighbor Summary
================================================

Interface  Down Attempt Init TwoWay ExStart Exchange Loading Full  Total
-------------------------------------------------------------------------
1/1/1      0    0       0    0      0       0        0       1     1
1/1/2      0    0       0    0      0       0        0       0     0
Total      0    0       0    0      0       0        0       1     1
```

show ip ospf routes

Syntax

```
show ip ospf [<PROCESS-ID>] routes
     [<IPV4-ADDR>/<MASK>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays OSPF routing table information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display information on the OSPF routing table for the particular OSPF process. Range: 1 to 63.

<IPV4-ADDR>

Specify an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. .

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32.

all-vrfs

Select to display OSPF routing table information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPF routing table information:

```
switch# show ip ospf routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table
---------------------------------------------

Total Number of Routes : 2

10.1.1.0/24        (i) area: 0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
20.1.1.0/24        (I)
     via 10.1.1.2 interface 1/1/1, cost 2 distance 110
```

Showing OSPF routing table information for a specific subnet:

```
switch# show ip ospf routes 10.1.1.0/2
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table for prefixes 10.1.1.0/24
----------------------------------------------------------------------

Total Number of Routes : 1
```

```
10.1.1.0/24          (i) area: 0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
```

show ip ospf statistics

Syntax

show ip ospf [*<PROCESS-ID>*] statistics [all-vrfs | vrf *<VRF-NAME>*]

Description

Displays OSPF statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF Process ID to display information on the OSPF statistics for the particular OSPF process. Range: 1 to 63.

all-vrfs

Select to display OSPF statistics information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPF statistics:

```
switch# show ip ospf statistics
OSPF Process ID 1 VRF default, Statistics (cleared 1h 16m 24s ago)
-------------------------------------------------------------

Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad Instance ID Drops            : 0
Bad IP Header Length Drops       : 0
Wrong OSPF Version Drops         : 0
Bad Source IP Drops              : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
```

show ip ospf statistics interface

Syntax

show ip ospf [*<PROCESS-ID>*] statistics interface [*<INTERFACE-NAME>*]
     [all-vrfs | vrf *<VRF-NAME>*]

Description

Displays OSPF statistics for the OSPF-enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Enter an OSPF process ID to display OSPF-enabled interface statistics information on the specified OSPF process. Range: 1 to 63.

<INTERFACE-NAME>

Select to display information only for the specified interface.

all-vrfs

Select to display OSPF-enabled interface statistics information for all VRFs.

vrf *<VRF-NAME>*

Specify the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPF-enabled interfaces information:

```
switch# show ip ospf statistics interface 1/1/1
OSPF Process ID 1 VRF default, interface 1/1/1 statistics (cleared 0h 30m 28s ago)
=============================================================================

Tx Hello Packets      : 101             Rx Hello Packets       : 99
Tx Hello Bytes        : 101             Rx Hello Bytes         : 99
Tx DD Packets         : 101             Rx DD Packets          : 99
Tx DD Bytes           : 101             Rx DD Bytes            : 99
Tx LS Request Packets : 101             Rx LS Requests Packets : 99
Tx LS Request Bytes   : 101             Rx LS Request Bytes    : 99
Tx LS Update Packets  : 101             Rx LS Update Packets   : 99
Tx LS Update Bytes    : 101             Rx LS Update Bytes     : 99
Tx LS Ack Packets     : 101             Rx LS Ack Packets      : 99
Tx LS Ack Bytes       : 101             Rx LS Ack Bytes        : 99

Total Number of State Changes : 8
Number of LSAs                : 29
LSA Checksum Sum              : 2345
Total Transmit Failures       : 29
Total OSPF Packets Discarded  : 999

Reason                               Packets Dropped
-----------------------------------------------------
Invalid type                         19
Invalid length                       9
Invalid checksum                     0
Invalid version                      23
Bad or unknown source                67
Area mismatch                        1
Self-originated                      19
```

```
Duplicate router ID                      9
Interface standby                        0
Total Hello packets dropped              60
  Network Mask mismatch                  10
  Hello interval mismatch                10
  Dead interval mismatch                 10
  Options mismatch                       10
  MTU mismatch                           10
  Neighbor ignored                       10
Authentication errors                    12
  Type mismatch                          6
  Authentication failures                6
Wrong protocol                           0
Resource failures                        0
Bad LSA length                           0
Others                                   0

Total LSAs Ignored : 176
Bad Type           : 10
Bad Length         : 56
Invalid Data       : 55
Invalid Checksum   : 55
```

summary-address

Syntax

`summary-address  <IPV4-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]`

`no summary-address <prefix/length> [no-advertise | tag <tag-value>`

Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The `no` form of this command disables route summarization.

NOTE: This command only works for an ASBR (Autonomous System Boundary Router).

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<IPV4-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32.

`no-advertise`

Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.

`tag` ***<TAG-VALUE>***

Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# summary-address 10.1.0.0/16
```

Disabling route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no summary-address 10.1.0.0/16
```

timers lsa-arrival

Syntax

```
timers lsa-arrival <DELAY>

no timers lsa-arrival
```

Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command `timers throttle lsa start` on the neighbor.

The `no` form of this command sets the LSA timers to default values.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<DELAY>

Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers lsa-arrival
```

timers throttle lsa

Syntax

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>

no timers throttle lsa
```

Description

Configures the timers for LSA generation.

The `no` form of this command sets the LSA timers to default values.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

`start-time <START-TIME>`

Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.

`hold-time <HOLD-TIME>`

Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until `max-wait-time` is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

`max-wait-time <WAIT-TIME>`

Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LSA timers:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers throttle lsa
```

timers throttle spf

Syntax

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TINME>
max-wait-time <WAIT-TIME>

no timers throttle spf
```

Description

Configures timers for SPF calculation. There are three timers:

- `start-time` Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- `hold-time` Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- `max-wait-time` Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The `no` form of this command sets all the configured non-default timers to default value.

Command context

`config-ospf-<PROCESS-ID>`

Parameters

<START-TIME>

Time in milliseconds to set timer for initial SPF delay. Default: 200.

<HOLD-TINME>

Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000.

<WAIT-TIME>

Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time 3000 max-wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
        area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

trap-enable

Syntax

`trap-enable`

`no trap-enable`

Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

The `no` form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no trap-enable
```

Overview

NOTE: OSPFv3 is not supported on the Aruba 6100 Switch Series.

OSPFv3 is the IPv6 implementation of Open Shortest Path First protocol (OSPFv2 is the IPv4 implementation of this protocol). OSPFv3 is a routing protocol which is described in RFC5340 entitled OSPF for IPv6. It is a link-state based IGP (Interior Gateway Protocol) routing protocol. It is widely used with medium to large-sized enterprise networks.

The characteristics of OSPFv3 are:

- Provides a loop-free topology using SPF algorithm.
- Allows hierarchical routing using area 0 (backbone area) as the top of the hierarchy.
- Supports load balancing with equal cost routes for same destination.
- OSPFv3 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Scales enterprise size network easily with area concept.
- Provides fast convergence with triggered, incremental updates through Link State Advertisements (LSAs).

Some OSPFv3 configuration is done in the configuration context, others in the OSPFv3 router context, or in the interface context. OSPFv3 can be configured on routed ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv3 mandates the associated interface to be routing interface.

The switch supports congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling).

Supported features

- OSPFv3 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling

- SPF throttling
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)

How OSPFv3 protocol works

OSPFv3 protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

OSPFv3 concepts

OSPFv3 is a link-state routing protocol applied to IPv6 routers grouped into OSPFv3 areas identified by the IPv6 routing configuration on each routing switch. Each OSPFv3 area includes one or more networks. OSPFv3 routers use hello packets and LSAs to maintain OSPFv3 operation across networks within an area and between areas within an OSPFv3 domain.

OSPFv3 uses hello packets to initiate and preserve relationships between neighboring routers on the same interface.

OSPFv3 Link-state advertisement (LSA) types

OSPFv3 uses LSAs transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces.

Each routing switch in an area also maintains an LSDB that describes the area topology. (All routers in a given OSPFv3 area have identical LSDBs, and each router uses the LSDB to build its own shortest-path tree.)

The routing switches used to connect areas to each other flood inter-area-prefix-LSAs, inter-area-router-LSAs, and AS-external-LSAs to backbone area to update it regarding available routes. Each OSPFv3 router determines the shortest path between itself and a desired destination router in the same OSPFv3 domain (AS). Routed traffic in an OSPFv3 AS is classified as one of the following:

- Intra-area traffic
- Inter-area traffic
- External traffic

The routing switches support the LSAs listed in the following table.

Link-state type	Description	Use	Flood scope
0x2001	Router-LSA	Describes the state of each active interface on a router for Area a given area. (Excludes loopback interfaces and interfaces that have not achieved full adjacency.)	Area
0x2002	Network-LSA	Describes the OSPFv3 routers in a given network.	Area
0x2003	Inter-area-prefix-LSA	Describes the route to a prefix in another OSPFv3 area of Area the same AS. (Excludes prefixes for link-local addresses.) Propagated through backbone area to other areas.	Area
0x2004	Inter-area-router-LSA	Describes the route to an ASBR in another OSPFv3 normal area (including the backbone area) of the same AS. Propagated through backbone area to other areas. (Excludes any ASBR in the same area as the router sending the LSA.)	Area
0x4005	AS-external-LSA	Describes the route to a destination prefix in another AS (external route). (Excludes prefixes for link-local addresses.) Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas. Does not flood over virtual links and is not summarized in virtual links. For injection into an NSSA, an NSSA ABR generates a type-7-default-LSA advertising the default route (::/0).	AS
0x2007	NSSA-LSA	Describes the route to a destination in another AS (external route). Originated by ASBR in NSSA. ABR translates type-7 LSAs to AS-external-LSAs for injection into the backbone area.	NSSA

Table Continued

Link-state type	Description	Use	Flood scope
0x0008	Link-LSA	For other routers on the same VLAN interface (except virtual links), describes the router link-local address and any other IPv6 prefixes reachable on the VLAN. Link LSAs are not flooded over virtual links.	Link-local
0x2009	Intra-area-prefix-LSA	Generated on transit links within an area by the DR operating on those links. Also, every OSPFv3 router generates this LSA to refer to stub and loopback prefixes on the router.	Area

OSPFv3 area types

OSPFv3 is built upon a hierarchy of network areas. All areas for a given OSPFv3 domain reside in the same AS. An AS is defined as a number of contiguous networks that share an interior gateway routing protocol.

An AS can be divided into multiple areas, including the backbone (area 0). Because each area represents a collection of contiguous networks and hosts, the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which router-LSAs and network-LSAs are broadcast. This functionality limits the amount of LSA flooding that occurs within the AS. It also helps to control the size of the link-state databases (LSDBs) maintained in OSPFv3 routers.

An area is represented in OSPFv3 by either a 32-bit dotted-decimal address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and Stub.

Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured, virtual link. The backbone is a special type of area and serves as a transit area for carrying the inter-area-prefix-LSAs, AS-external-LSAs, and routed traffic between non-backbone areas, as well as the router-LSAs, network-LSAs, and routed traffic internal to the area. ASBRs are allowed in backbone areas.

Normal area

A normal area allows inter-area-prefix-LSAs and AS-external-LSAs to and from the backbone area. A normal area connects to the AS backbone area through one or more ABRs (physically or through a virtual link) and allows router-LSAs and network LSAs within this area. ASBRs are allowed in normal areas.

Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow AS-external-LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise inter-area routes from other areas.
- Use the inter-area-prefix-LSA default route to advertise routes to an ASBR and to other areas.

You can configure the stub area ABR to do the following:

- Suppress advertising on the area's summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing inter-area-prefix-LSAs and the default external route from the backbone area with the default summary route (::/0.). This area is called totally stubby area.

Virtual links are not allowed for stub areas.

Not-so-stubby (NSSA) area

An NSSA area connects to the backbone area through one or more ABRs. NSSAs are used where an ASBR exists in an area where you want to:

- Block injection of external routes from other areas of the AS.
- Advertise type-7-LSA external routes (learned from the ASBR) to the backbone area as AS-external-LSAs.

NSSAs also support the following:

- Advertise inter-area-prefix-LSAs from the backbone area into the NSSA. (If no-summary is enabled, the NSSA ABR suppresses these LSAs from the backbone and, instead, injects the default route into the NSSA.)
- Advertise NSSA inter-area-prefix-LSAs to the backbone area.

Virtual links are not allowed for NSSAs.

OSPFv3 router types

Internal routers

Internal OSPFv3 routers belong to only one area. Internal routers flood router-LSAs to all routers in the same area and maintain identical LSDBs.

Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding inter-area-prefix-LSAs and inter-area router LSAs between its border areas.

You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IPv6 addresses. This aggregate address is advertised instead of all the individual addresses it represents. You can assign up to eight ranges in an OSPFv3 area.

Autonomous system boundary router (ASBR)

Autonomous system boundary routers run one or more interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPFv3 through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

Designated routers (DRs)

In an OSPFv3 network having two or more routers, one router is elected to serve as the designated router (DR) and another router to act as the backup designated router (BDR). All other routers in the same network segment forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This functionality minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPFv3 network with no DR and no BDR, the neighboring router with the highest priority is elected the DR and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPFv3 network are declaring themselves DRs, both priority and router ID are used to select the DR and BDR.

Priority is configurable using the `ipv6 ospfv3 priority` command at the interface level. You can use this command to help bias one router as the DR. If two neighbors share the same priority, the router with the highest router ID is designated the DR. The router with the next highest router ID is designated the BDR. Routers retain DR/BDR states throughout the period for which the adjacency is up and running.

OSPFv3 configuration task list

Tasks at a glance

- **Configuring OSPFv3 on the routing switch**
- **Assigning the routing switch to an OSPF area**
- **Setting OSPFv3 network for the area**
- **Configuring external route redistribution and control**
- **Influencing route choice by changing the administrative distance**
- **Configuring graceful restart**
- **Configuring OSPFv3 interface settings**
- **Configuring all OSPFv3 interfaces as passive**
- **Viewing OSPFv3 information**
- **Clearing OSPFv3 statistics on a switch**

Configuring OSPFv3 on the routing switch

Create the OSPFv3 instance and enter the OSPFv3 configuration context. From this context, you can proceed with other OSPFv3 configuration.

Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPFv3 instance and enter the OSPFv3 configuration context.
- To configure a router ID, and other OSPFv3 configuration you must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1 )#` prompt.

Procedure

1. Create the OSPFv3 instance and enter the OSPFv3 configuration context using the following command. For command details, see **`router ospfv3`**.

```
router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

For example, the following command creates OSPF instance 1.

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

2. Configure a global router ID using the following command. For command details, see **router-id**.

```
router-id <ROUTER-ADDRESSW>
```

For example, the following command sets the router ID to 1.1.1.1.

```
switch(config-ospfv3-1)# router-id 1.1.1.1
```

3. Optionally, if the OSPFv3 process was disabled (it is enabled by default), enable the OSPFv3 process using the following command.

```
enable
```

For command details, see **enable**. (Refer to **disable** for disabling the OSPFv3 process).

Creating an OSPFv3 area

Create a Normal, Stub, or Not So Stubby (NSSA) area.

Prerequisites

You must be in the OSPFv3 router configuration context.

Procedure

1. Create an OSPFv3 area for the routing switch using one of the following commands:

   - Create a Normal area using the following command: `area <area-id>`. For command details, see .
   - Create a Stub area using the following command: `area <area-id> stub`. For command details, see .
   - Create a Not So Stubby Area using the command: `area <area-id> nssa`.

   For example, the following command creates a normal area with an area identifier of 10.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospfv3-1)# area 1.1.1.1
```

Setting OSPFv3 network for the area

After you define an OSPFv3 area, you can assign one or more networks to it. The OSPFv3 protocol will run on the interface with the configured link-local ipv6 address. The interfaces that have an IPv6 address configured in this network or in a subset of this network will participate in the OSPFv3 protocol.

Prerequisites

Routing must be enabled on the interface where you are planning to enable OSPFv3.

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

1. Set an OSPFv3 network for the area using the following command. For command details, see **ipv6 ospfv3 area**.

   ```
   ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
   ```

   For example, use the following command to assign interface 1/1/1 to OSPFv3 area 1. The area can alternatively be entered in IO address format.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ipv6 ospfv3 1 area 1
   ```

2. Optionally, you can disable OSPFv3 on the interface using the following command. For command details, see **ipv6 ospfv3 shutdown**.

   ```
   ipv6 ospfv3 shutdown
   ```

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ipv6 ospfv3 shutdown
   ```

Configuring external route redistribution and control

Configuring route redistribution for OSPFv3 establishes the routing switch as an ASBR for importing and translating different protocol routes from other IGP domains into an OSPFv3 domain. When you configure redistribution for OSPFv3, you can specify that static, or connected routes external to the OSPFv3 domain are imported as OSPFv3 routes.

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

Procedure

1. Enable route redistribution using the following command and specify one of the following parameters: connected and static. For command details, see **redistribute**.

   ```
   redistribute {connected | static | ripng | ospfv3 <PROCESS-ID>} [route-map
   <ROUTE-MAP-NAME>]
   ```

   For example, the following command sets redistribution of connected routes as OSPF routes:

   ```
   switch(config-ospfv3-1)# redistribute connected
   ```

2. Optionally, modify the default metric for redistribution using the following command. For command details, see **default-metric**.

   ```
   default-metric <METRIC-VALUE>
   ```

   For example, the following command sets the default metric for redistribution to 37.

   ```
   switch(config-ospfv3-1)# default-metric 37
   ```

3. Optionally, set the cost of default-summary LSAs using the following command. For command details, see .

   ```
   area <AREA-ID> default-metric <METRIC>
   ```

   For example, the following command sets the cost of default summary LSAs to 2.

```
switch(config-ospfv3-1)# area 1 default-metric 2
```

4. Optionally, set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. For command details, see **max-metric router-lsa**.

   ```
   max-metric router-lsa [on-startup [<INTERVAL>]]
   ```

   For example, the following command sets advertise max-metric router-lsa on startup.

   ```
   switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
   ```

Influencing route choice by changing the administrative distance

The administrative distance is used to influence the selection of routes learned by different protocols.

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

Procedure

Reconfigure the administrative distance using the following command. For command details, see **distance**.

```
distance <distance>
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospfv3-1)# distance 100
```

Configuring graceful restart

OSPFv3 routing can be gracefully restarted on switches without losing packets that are in transit. There is no effect on the saved switch configuration.

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

Procedure

Configure graceful restart of using the following command. For command details, see **graceful-restart**.

```
graceful-restart [restart-interval <seconds> | helper]
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospfv3-1)# graceful-restart restart-interval 50
```

Configuring OSPFv3 interface settings

You can optionally adjust the following OSPFv3 interface settings.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

- Set the interface cost using the following command. For command details, see **ipv6 ospfv3 cost**.

  `ipv6 ospfv3 cost <interface-cost>`

  For example, the following command sets the cost associated with interface 1/1/1 to 100.

  ```
  switch(config)# interface 1/1/1
  switch(config-if)# ipv6 ospfv3 cost 100
  ```

- Set the time interval between OSPFv3 hello packets for the OSPFv3 interface using the following command. For command details, see **ipv6 ospfv3 hello-interval**.

  `ipv6 ospfv3 hello-interval <seconds>`

- Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 interface using the following command. For command details, see **ipv6 ospfv3 dead-interval**.

  `ipv6 ospfv3 dead-interval <seconds>`

- Set the time between re-transmitting lost link state advertisements for the OSPFv3 interface using the following command. For command details, see **ipv6 ospfv3 retransmit-interval**.

  `ipv6 ospfv3 retransmit-interval <seconds>`

- Sets the time delay in Link state transmission for the OSPFv3 interface using the following command. For command details, see **ipv6 ospfv3 transit-delay**.

  `ipv6 ospfv3 transit-delay <seconds>`

- Set the OSPFv3 network type for the interface. For command details, see **ipv6 ospfv3 network**.

  `ipv6 ospfv3 network {broadcast|point-to-point}`

- Set the OSPFv3 priority on the interface using the following command. For command details, see **ipv6 ospfv3 priority**.

  `ipv6 ospfv3 priority <number-value>`

- Set the OSPFv3 interface as OSPFv3 passive interface. With this setting the interface participates in OSPFv3 but does not send or receive packets on that interface. For command details, see **ipv6 ospfv3 passive**.

  `ipv6 ospfv3 passive`

Configuring all OSPFv3 interfaces as passive

OSPFv3 sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPFv3 to be passive.

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

Procedure

Configure all OSPFv3 interfaces as passive using the following command. For command details, see **passive-interface default**.

```
passive-interface default
```

```
switch(config-ospfv3-1)# passive-interface default
```

Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

Prerequisites
You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure SPF throttling timers using the following command. For command details, see **timers throttle spf**

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-
time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time
3000 max-wait-time 9000
```

Viewing OSPFv3 information

Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

Procedure

1. To view OSPFv3 information, use the following commands. For command details and examples, click the links.
   - To view general OSPFv3, area, state and configuration information use: **show ipv6 ospfv3**.
   - To view information about OSPFv3 interfaces use: **show ipv6 ospfv3 interface**.
   - To view information about OSPFv3 neighbors use: **show ipv6 ospfv3 neighbors**.
   - To view OSPFv3 routing table information use: **show ipv6 ospfv3 routes**.
   - To view OSPFv3 statistics for the OSPFv3 interfaces use **show ipv6 ospfv3 statistics interface**.

Clearing OSPFv3 statistics on a switch

Clear OSPFv3 event statics using the following command. For command details, see **clear ipv6 ospfv3 statistics**

clear ipv6 ospfv3 [*<PROCESS-ID>*] statistics [interface [*<INTERFACE-NAME>*]] [all-vrfs | vrf *<VRF-NAME>*]

For example, the following command clears OSPFv3 event statistics from interface 1/1/1.

```
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

OSPFv3 commands

area

Syntax

area *<AREA-ID>*

no area *<AREA-ID>*

Description

Creates a normal area with <AREA-ID> set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The no form of this command deletes the area with the <AREA-ID> specified. The area can be of any type (stub, stub no-summary, and default normal area).

Command context

config-ospfv3-*<PROCESS-ID>*

Parameters

<AREA-ID>

Specifies the area ID in one of the following formats.

- Area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- Area identifier in decimal format. Range: 0 to 4294967295.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a normal area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1
```

Deleting an area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1
```

area authentication ipsec

Syntax

```
area <AREA-ID> authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>

no area <AREA-ID> authentication
```

Description

Configures IPsec AH authentication for the specified area. OSPFv3 interfaces which have IPsec configured at the interface context will not use area level IPsec.

The `no` form of this command removes IPsec AH authentication for the specified area.

NOTE: IPsec is not supported for 6in6 tunnel interfaces.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<AREA-ID>

Specifies the area ID is one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

spi *<SPI-INDEX>*

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256-4294967295 characters.

<AUTH-TYPE>

Specifies the algorithm to use for authentication: `md5` or `sha1`.

<KEY-TYPE>

Specifies the key type to use: `plaintext` (unencrypted), `hex-string` (encrypted) or `ciphertext` (encrypted).

<AUTH-KEY>

Specifies the key.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting area 1 to use IPsec authentication(AH):

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1 authentication ipsec spi 256 sha1 plaintext abcd
```

Removing IPsec authentication (AH) on area 1:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1 authentication
```

area encryption ipsec

Syntax

```
area <AREA-ID> encryption ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>
({<ENCR-TYPE> <KEY-TYPE> <ENCR-KEY>} | NULL)
```

```
no area <AREA-ID> encryption
```

Description

Configures IPsec ESP with authentication/encryption algorithm type and key for the specified area. OSPFv3 interfaces with IPsec configured at interface context will not use area level IPsec ESP configuration.

The `no` form of this command removes IPsec ESP from the specified area.

NOTE: IPsec is not supported for 6in6 tunnel interfaces.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<AREA-ID>

Specifies the area ID is one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

spi *<SPI-INDEX>*

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295 characters.

<AUTH-TYPE>

Specifies the algorithm to use for authentication: `md5` or `sha1`.

<ENCR-TYPE>

Specifies the algorithm to use for encryption: `3des`, `aes`, `des` or `null`.

NOTE: `aes` will be considered AES128, AES192 or AES256 based on key length.

<KEY-TYPE>

Specifies the key type to use: `plaintext` (unencrypted) `hex-string` (encrypted) or `ciphertext` (encrypted).

<AUTH-KEY>

Specifies authentication key.

<ENCR-KEY>

Specifies encryption key.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting area 0 to use IPSec ESP:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext abcd des plaintext abcdefab
```

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext abcd null
```

Removing IPSec ESP on area 0:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 0 encryption
```

clear ipv6 ospfv3 neighbors

Syntax

```
clear ipv6 ospfv3 [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Resets the neighbor and clears the OSPF neighbor information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 63.

<NEIGHBOR>

Specifies the router ID of a neighbor.

<INTERFACE-NAME>

Specifies the OSPFv3 statistics to clear for the specified interface.

all-vrfs

Select to clear the OSPFv3 statistics for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Clearing the OSPFv3 neighbor information:

```
switch# clear ipv6 ospfv3 1 neighbor 3.3.3.3
switch# clear ipv6 ospfv3 1 neighbor interface 1/1/1
```

clear ipv6 ospfv3 statistics

Syntax

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Clears the OSPFv3 event statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 63.

<INTERFACE-NAME>

Specifies the OSPFv3 statistics to clear for the specified interface.

all-vrfs

Select to clear the OSPFv3 statistics for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Clearing the OSPFv3 event statistics:

```
switch# clear ipv6 ospfv3 statistics
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

default-metric

Syntax

```
default-metric <METRIC-VALUE>
```

```
no default-metric
```

Description

Sets the default metric for redistributed routes in the OSPFv3.

The `no` form of this command sets the default metric to be used for redistributed routes into OSPFv3 to the default of 25.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<METRIC-VALUE>

Specifies the default metric value to use for redistributed routes. Range: 1 to 1677214. Default: 25.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-metric
```

disable

Syntax

`disable`

Description

Disables the OSPFv3 process. By default OSPFv3 process is enabled.

IMPORTANT: This command does not remove the OSPFv3 configurations.

Command context

`config-ospfv3-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Example

Disabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# disable
```

distance

Syntax

`distance <DISTANCE>`

`no distance`

Description

Defines an administrative distance for OSPFv3. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The `no` form of this command sets the OSPFv3 administrative distance to the default of 110.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<DISTANCE>

Specifies OSPFv3 administrative distance. Range: 1 to 255. Default: 110.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF administrative distance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distance 100
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distance
```

enable

Syntax

`enable`

Description

Enables OSPFv3 process when disabled. By default OSPFv3 process is enabled.

Command context

`config-ospfv3-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# enable
```

default-information originate

Syntax

`default-information originate`

`no default-information originate`

Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table.

The `no` form of this command disables advertisement of the default route.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

default-information originate always

Syntax

`default-information originate always`

`no default-information originate always`

Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

Command context

`config-ospf-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

graceful-restart

Syntax

`graceful-restart {restart-interval <INTERVAL> | helper | strict-lsa-check}`

`no graceful-restart {restart-interval | helper | strict-lsa-check}`

Description

Configures graceful restart for OSPFv3. By default graceful restart is enabled on the OSPFv3 router.

The `no` form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

`restart-interval <INTERVAL>`

Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120.

`helper`

Specifies that the router will participate in the graceful restart of a neighbor router.

`strict-lsa-check`

Enable strict LSA checking when acting as a restart helper for a restarting peer.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# graceful-restart restart-interval 40
switch(config-ospfv3-1)# graceful-restart helper
switch(config-ospfv3-1)# graceful-restart strict-lsa-check
```

Setting the restart interval to default, disabling helper mode, and strict LSA checking:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
switch(config-ospfv3-1)# no graceful-restart strict-lsa-check
```

ipv6 ospfv3 area

Syntax

```
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>

no ipv6 ospfv3 <PROCESS-ID> area <area-id>
```

Description

Runs the OSPFv3 protocol on the interface for the area specified.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The `no` form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

Command context

`config-if`

`config-if-vlan`

Parameters

<PROCESS-ID>

Specifies the OSPFv3 process ID. Range: 1 to 63.

<AREA-ID>

Specifies the area ID in one of the following formats.

- Area ID in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- Area ID as a decimal value. Range: 0-4294967295.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 0.0.0.1
```

Disabling OSPFv3 network for the area:

```
switch(config)# interface 1/1/1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 0.0.0.1
```

ipv6 ospfv3 authentication null

Syntax

```
ipv6 ospfv3 authentication null
```

Description

Configures null authentication on an interface which disables IPsec authentication.

Command context

`config-if`

`config-if-vlan`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling IPsec on interface VLAN **1**:

```
switch(config)# interface van 1
switch(config-if-vlan)# ipv6 ospfv3 authentication null
```

ipv6 ospfv3 authentication ipsec

Syntax

`ipv6 ospfv3 authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>`

`no ipv6 ospfv3 authentication`

Description

Configures IPSec AH authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec.

The `no` form of this command removes IPsec AH authentication for the specified area.

Command context

`config-if`

`config-if-vlan`

Parameters

spi *<SPI-INDEX>*

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256-4294967295 characters.

<AUTH-TYPE>

Specifies the algorithm to use for authentication: `md5` or `sha1`.

<KEY-TYPE>

Specifies the key type to use: `plaintext` (not encrypted) or `hex-string` (encrypted) or `ciphertext` (encrypted).

<AUTH-KEY>

Specifies the key.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting interface VLAN **1** to use IPsec authentication (AH):

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5 plaintext abcd
```

Removing IPsec authentication (AH) on interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 authentication
```

ipv6 ospfv3 cost

Syntax

```
ipv6 ospfv3 cost <INTERFACE-COST>

no ipv6 ospfv3 cost
```

Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The `no` form of this command sets the cost (metric) associated with a particular interface to the default of 1.

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERFACE-COST>

Specifies the interface cost value. Range: 1to 65535. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 interface cost:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 cost 100
```

Setting the OSPFv3 interface cost to default:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 cost
```

ipv6 ospfv3 dead-interval

Syntax

```
ipv6 ospfv3 dead-interval <INTERVAL>

no ipv6 ospfv3 dead-interval
```

Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 interface.

The `no` form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 interface. The default value is 40 seconds (generally four times the hello packet interval).

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERVAL>

Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 dead-interval 30
```

Setting OSPFv3 dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 dead-interval
```

ipv6 ospfv3 encryption ipsec

Syntax

ipv6 ospfv3 encryption ipsec spi *<SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>*
({*<ENCR-TYPE> <KEY-TYPE> <ENCR-KEY>*} | null)

no ipv6 ospfv3 encryption

Description

Configures IPsec ESP. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The `no` form of this command removes IPsec ESP for the specified area.

Command context

`config-if`

`config-if-vlan`

Parameters

spi *<SPI-INDEX>*

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec ESP header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295 characters.

<AUTH-TYPE>

Specifies the algorithm to use for authentication: `md5` or `sha1`.

<ENCR-TYPE>

Specifies the algorithm to use for encryption: `des`, `3des` or `aes`.

<KEY-TYPE>

Specifies the key type to use: `plaintext` (not encrypted), `hex-string` (encrypted) or `ciphertext` (encrypted).

<AUTH-KEY>

Specifies the authentication key.

<ENCR-KEY>

Specifies the encryption key.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting interface VLAN **1** to use IPsec ESP:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext abcdef aes
plaintext abcdefabcdefabcd
```

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext abcdef null
```

Removing IPsec on interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 encryption
```

ipv6 ospfv3 encryption null

Syntax

`ipv6 ospfv3 encryption null`

Description

Configures NULL ESP on an interface which disables IPsec ESP.

Command context

`config-if`

`config-if-vlan`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disable IPsec ESP on interface VLAN *1*:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption null
```

ipv6 ospfv3 hello-interval

Syntax

`ipv6 ospfv3 hello-interval <INTERVAL>`

`no ipv6 ospfv3 hello-interval`

Description

Sets the time interval between OSPFv3 hello packets for the OSPFv3 interface.

The `no` form of this command sets the time interval between OSPFv3 hello packets to the default for the OSPFv3 interface of 10 seconds.

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERVAL>

Specifies the time interval between hello packets, in seconds. Range: 1 to 65535. Default: 10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 hello-interval 30
```

Setting OSPFv3 hello interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 hello-interval
```

ipv6 ospfv3 network

Syntax

`ipv6 ospfv3 network {broadcast|point-to-point}`

`no ipv6 ospfv3 network`

Description

Configures the network type for the interface. By default the network type is broadcast network.

The `no` form of this command sets the network type for the interface to the system default of broadcast network.

Command context

`config-if`

`config-if-vlan`

Parameters

`broadcast`

Specifies the OSPFv3 network type as a broadcast multiaccess network.

`point-to-point`

Specifies the OSPFv3 network type as a point-to-point network.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 network broadcast
switch(config-if-vlan)# ipv6 ospfv3 network point-to-point
```

Disabling OSPFv3 network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 network
```

ipv6 ospfv3 passive

Syntax

```
ipv6 ospfv3 passive

no ipv6 ospfv3 passive
```

Description

Configures the interface as an OSPFv3 passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The `no` form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

Command context

`config-if`

`config-if-vlan`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the interface as OSPFv3 passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 passive
```

Setting the interface as OSPFv3 active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 passive
```

ipv6 ospfv3 priority

Syntax

```
ipv6 ospfv3 priority <number-value>

no ipv6 ospfv3 priority
```

Description

Sets the OSPFv3 priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The `no` form of this command sets the OSPFv3 priority for the interface to the default of 1.

Command context

`config-if`

`config-if-vlan`

Parameters

<number-value>

Specifies the OSPFv3 priority value. Default: 1. Range: 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the OSPFv3 priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if-vlan)# ipv6 ospfv3 priority 50
```

Setting the OSPFv3 priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 priority
```

ipv6 ospfv3 retransmit-interval

Syntax

```
ipv6 ospfv3 retransmit-interval <INTERVAL>

no ipv6 ospfv3 retransmit-interval
```

Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 interface.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default 5 seconds.

Command context

`config-if`

`config-if-vlan`

Parameters

<INTERVAL>

Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 retransmit-interval 30
```

Setting OSPFv3 retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 retransmit-interval
```

ipv6 ospfv3 shutdown

Syntax

```
ipv6 ospfv3 shutdown

no ipv6 ospfv3 shutdown
```

Description

Disables OSPFv3 on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command `no ip ospf area`.

The `no` form of this command re-enables OSPFv3 on the interface.

Command context

`config-if`

`config-if-vlan`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 shutdown
```

Re-enabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 shutdown
```

ipv6 ospfv3 transit-delay

Syntax

```
ipv6 ospfv3 transit-delay <DELAY>
```

```
no ipv6 ospfv3 transit-delay
```

Description

Sets the time delay in Link state transmission for the OSPFv3 interface.

The `no` form of this command sets the transit delay in Link state transmission to the default of 1 second.

Command context

`config-if`

`config-if-vlan`

Parameters

<DELAY>

Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 transit delay on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 transit-delay 30
```

Setting OSPFv3 transit delay to default on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 transit-delay
```

maximum-paths

Syntax

```
maximum-paths <MAXIMUM>
```

```
no maximum-paths
```

Description

Sets the maximum number of ECMP routes that OSPFv3 can support.

The `no` form of this command sets the maximum number of ECMP routes that OSPFv3 can support to the default value of 4.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<MAXIMUM>

Specifies the maximum number of ECMP routes. Range: 1 to 8. Default: 4.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# maximum-paths 3
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no maximum-paths
```

max-metric router-lsa

Syntax

`max-metric router-lsa [on-startup <INTERVAL>]`

`no max-metric router-lsa [on-startup]`

Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup. That is, for the time specified in seconds, or the default value of 600 seconds.

To disable advertisement of the maximum metric, use the `no` form of the command.

The `no` form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

on-startup *<INTERVAL>*

Automatically advertises the stub Router-LSA (or maximize the router-LSA cost metric) for a specified time interval upon OSPFv3 startup.

Specifies the time in seconds. Range: 5 to 86400. Default: 600.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# max-metric router-lsa
switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no max-metric router-lsa
switch(config-ospfv3-1)# no max-metric router-lsa on-startup
```

passive-interface default

Syntax

`passive-interface default`

`no passive-interface default`

Description

Sets all OSPFv3 interfaces as passive.

The `no` form of this command sets all the OSPFv3 interfaces as active.

Command context

`config-ospfv3-<PROCESS-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3-enabled interfaces as passive:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# passive-interface default
```

Setting OSPFv3-enabled interfaces as active:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no passive-interface default
```

redistribute

Syntax

`redistribute {connected | static | ripng | ospfv3 <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]`

`no redistribute {connected | static} [route-map <ROUTE-MAP-NAME>]`

Description

Redistributes routes originating from other protocols, or from another OSPF process, to the current OSPF process.

The `no` form of this command disables redistribution of routes to the current OSPF process.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

`connected`

Specifies redistributing connected routes (directly attached subnet or host) to OSPFv3.

`static`

Specifies redistributing static routes to OSPFv3.

`ripng`

Specifies redistributing RIPng routes.

`ospfv3 <PROCESS-ID>`

Specifies redistributing routes from the specified OSPFv3 process ID.

`route-map <ROUTE-MAP-NAME>`

Specifies the name of a route map. To create a route map, use the `route-map` command.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
switch(config-ospfv3-1)# redistribute connected
switch(config-ospfv3-1)# redistribute static
switch(config-ospfv3-1)# redistribute ripng
switch(config-ospfv3-1)# redistribute oppfv3 1
switch(config-ospfv3-1)# redistribute route-map static_networks
```

Disabling redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
switch(config-ospfv3-1)# no redistribute connected
switch(config-ospfv3-1)# no redistribute static
switch(config-ospfv3-1)# no redistribute ripng
switch(config-ospfv3-1)# no redistribute oppfv3 1
switch(config-ospfv3-1)# no redistribute route-map static_networks
```

reference-bandwidth

Syntax

```
reference-bandwidth <BANDWIDTH>

no reference-bandwidth
```

Description

Sets the reference bandwidth for OSPFv3. If the OSPFv3 interface cost is not explicitly set, then the cost of all the OSPFv3 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 interface cost is not explicitly set).

The `no` form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<BANDWIDTH>

Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# routerv3 ospf 1
switch(config-ospfv3-1)# no reference-bandwidth
```

retransmit-interval

Syntax

`retransmit-interval <INTERVAL>`

`no retransmit-interval`

Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

Command context

`config-router-vlink`

Parameters

<INTERVAL>

Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 virtual links retransmit interval:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 virtual links retransmit interval to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
```

router-id

Syntax

`router-id <ROUTER-ADDRESS>`

`no router-id`

Description

Sets an ID for the router in an IPv4 address format.

The `no` form of this command unconfigures the router-id for the instance and sets the router-id to the default. The router-id is changed to the dynamically selected router-id. The default router-id 0.0.0.0 updates to the routing stack that triggers to auto-elect a router-id based on the highest IP address of loopback interface, or the highest IP address of interfaces.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<ROUTER-ADDRESS>

Specifies the router address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting router-id in the OSPFv3 context:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no router-id
```

router ospfv3

Syntax

`router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]`

`no router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]`

Description

Creates the OSPFv3 process (if not created already) and enters the router OSPFv3 instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the *<vrf-name>* is not specified. Only one OSPFv3 process is allowed per VRF.

The `no` form of this command removes the OSPFv3 instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <*var-name*> is not specified.

Command context

`config`

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID. Length: 1 to 63.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Entering the router OSPFv3 instance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3 1 vrf vrf_red
```

Removing the router OSPFv3 instance:

```
switch#(config)# no router ospfv3 1
```

show ipv6 ospfv3

Syntax

`show ipv6 ospfv3 [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]`

Description

Shows OSPFv3 information including area, state, and configuration information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID optionally to show OSPFv3 information for a particular OSPFv3 process. Range: 1 to 63.

all-vrfs

Select to show OSPFv3 information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing general OSPFv3 configurations:

```
switch# show ipv6 ospfv3 1
Routing Process 1 with ID: 1.1.1.1 VRF default
-----------------------------------------------

OSPFv3 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of External LSAs 0, checksum sum 0
Summary address:
  prefix fd00::0/64 advertise tag 10
Number of areas is: 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is: 1, 1 normal, 0 stub, 0 NSSA
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 9 times
  Number of LSAs: 4, checksum sum 195745
Area (0.0.0.1) (Inactive)
  Interfaces in this Area: 0 Active Interfaces: 0
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 8 times
  Area ranges:
    fd00::1/64, inter-area, advertise
  Number of LSAs: 0, checksum sum 0
```

show ipv6 ospfv3 border-routers

Syntax

`show ipv6 ospfv3 [<PROCESS-ID>] border-routers [all-vrfs | vrf <VRF-NAME>]`

Description

Shows the OSPFv3 routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID to show the OSPFv3 routing table entries for ABR and ASBR for the particular OSPFv3 process. Range: 1-63.

all-vrfs

Select to show OSPFv3 border router information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing OSPFv3 border routers information for default VRF:

```
switch# show ipv6 ospfv3 border-routers
OSPFv3 Process ID 1 VRF default, Internal Routing Table
-------------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 17 via
        fe80::98f2:b301:2c68:1d48, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 17 via
        fe80::98f2:b301:3068:1d48, Interface 1/1/2
```

Showing OSPFv3 border routers information for default VRF:

```
switch# show ipv6 ospfv3 border-routers
OSPFv3 Process ID 1 VRF default, Internal Routing Table
-------------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 17 via
        fe80::98f2:b301:2c68:1d48, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 17 via
        fe80::98f2:b301:3068:1d48, Interface 1/1/2
```

show ipv6 ospfv3 interface

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID optionally to show the OSPFv3 enabled interfaces for the particular OSPFv3 process. Range: 1-63.

<INTERFACE-NAME>

Selects to show information only for the specified OSPFv3 enabled interface.

brief

Select to show the overview information for the OSPFv3 enabled interface.

all-vrfs

Select to show OSPF-enabled interface information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPFv3 information for all interfaces in default VRF:

```
switch# show ipv6 ospfv3 interface
Interface 1/1/1 is up, Line Protocol is up
-------------------------------------------
IPv6 address fe80::98f2:b301:468:be02
Process ID 1 VRF default, Area 0.0.0.0
    State Dr, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit Delay 1 sec, Router Priority 1
    Designated Router ID: 1.1.1.1
    Backup Designated Router ID: 2.2.2.2
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    Number of Link LSAs: 2, checksum sum 67473

Interface 1/1/2 is up, Line Protocol is up
-------------------------------------------

IPv6 address fe80::98f2:b301:868:be02
Process ID 1 VRF default, Area 0.0.0.1
    State Waiting, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit Delay 1 sec, Router Priority 1
    No designated router on this network
    No backup designated router on this network
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    Number of Link LSAs: 1, checksum sum 24592
```

Showing overview information for OSPFv3 enabled interfaces for all VRFs:

```
switch# show ipv6 ospfv3 interface brief all-vrfs
OSPFv3 Process ID 1 VRF default
================================

```

```
Total Number of Interfaces: 2

Interface      Area                 Cost      State              Status
-----------------------------------------------------------------------
1/1/1          0.0.0.0              10        Dr                 up
1/1/2          0.0.0.1              10        Backup-dr          up
```

show ipv6 ospfv3 neighbors

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
     [interface <INTERFACE-NAME>] [detail | summary]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 neighbors.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID to show OSPFv3 neighbor information for the particular OSPFv3 process. Range: 1-63.

neighbors *<NEIGHBOR-ID>*

Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D).

interface *<INTERFACE-NAME>*

Shows neighbor information only for the specified interface.

detail

Shows detailed information for the neighbors.

summary

Shows summary information for the neighbors.

all-vrfs

Shows neighbor information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPFv3 neighbors information:

```
switch# show ipv6 ospfv3 neighbors
OSPFv3 Process ID 1 VRF default
================================
```

```
Total Number of Neighbors: 1

Neighbor ID      Priority  State             Interface
-------------------------------------------------------
2.2.2.2          1         FULL/BDR          1/1/1
  Neighbor address fe80::98f2:b301:2c68:1d48
```

Showing OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48
--------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    Neighbor priority is 1, State is FULL
    Options is 0x13
    Dead Timer due in 00:00:38
    Time since last state change 00h:20m:26s
```

Showing detail OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 detail
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48
--------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    Neighbor priority is 1, State is FULL
    DR is 1.1.1.1, BDR is 2.2.2.2
    Options is 0x13
    Dead Timer due in 00:00:36
    Retransmission Queue Length 0
    Time since last state change 00h:20m:38s
```

Showing OSPFv3 neighbors information for interface `1/1/1`:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 interface 1/1/1
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48
--------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    Neighbor priority is 1, State is FULL
    Options is 0x13
    Dead Timer due in 00:00:32
    Time since last state change 00h:20m:52s
```

show ipv6 ospfv3 routes

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] routes [<PREFIX/LENGTH>]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 routing table information.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID that shows information from the OSPFv3 routing table for the particular OSPFv3 process. Range: 1-63.

<PREFIX/LENGTH>

Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32.

all-vrfs

Select to show OSPFv3 routing table information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPFv3 routing table information:

```
switch# show ipv6 ospfv3 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table
-----------------------------------------------

Total Number of OSPFv3 Routes : 2

111::/64           (i) area:0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 routing table information for VRF red:

```
switch# show ipv6 ospfv3 routes vrf red

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 2 VRF red, Routing Table
-------------------------------------------

Total Number of OSPFv3 Routes : 1

222::/64           (i) area:0.0.0.1
     directly attached to interface 1/1/2, cost 1 distance 110
```

Showing OSPFv3 routing table information for destination prefix fd00::/64:

```
switch# show ipv6 ospfv3 1 routes fd00::/64
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table for prefixes fd00::/64
-----------------------------------------------------------------------
```

```
Total Number of OSPFv3 Routes : 1

fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

show ipv6 ospfv3 statistics

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows OSPFv3 statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID that shows information on the OSPFv3 SPF statistics for the particular OSPFv3 process. Range: 1-63.

all-vrfs

Select to show OSPFv3 SPF statistics information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing OSPFv3 statistics information:

```
switch# show ipv6 ospfv3 statistics
OSPFv3 Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
--------------------------------------------------------------------

Unknown Interface Drops         : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops    : 0
Wrong OSPFv3 Version Drops      : 0
Bad Source IPv6 Drops           : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
```

Showing OSPFv3 statistics information for VRF red:

```
switch# show ipv6 ospfv3 2 statistics vrf red

OSPFv3 Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
----------------------------------------------------------------

```

```
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
```

show ipv6 ospfv3 statistics interface

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 statistics for the OSPFv3-enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies an OSPFv3 process ID to show OSPF-enabled interface statistics information on the specified OSPFv3 process. Range: 1 to 63.

<INTERFACE-NAME>

Selects to show information only for the specified interface.

all-vrfs

Select to show OSPF-enabled interface statistics information for all VRFs.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing OSPFv3-enabled interfaces information for interface 1/1/1:

```
switch# show ipv6 ospfv3 statistics interface all-vrfs
OSPFv3 Process ID 1 VRF default, Interface vlan2000 Statistics  (cleared 0h 2m 52s ago)
======================================================================================

Tx Hello packets      : 0                    Rx Hello packets      : 0
Tx Hello bytes        : 0                    Rx Hello bytes        : 0
Tx DD packets         : 0                    Rx DD packets         : 0
Tx DD bytes           : 0                    Rx DD bytes           : 0
Tx LS request packets : 0                    Rx LS request packets : 0
Tx LS request bytes   : 0                    Rx LS request bytes   : 0
Tx LS update packets  : 0                    Rx LS update packets  : 0
Tx LS update bytes    : 0                    Rx LS update bytes    : 0
Tx LS ack packets     : 0                    Rx LS ack packets     : 0
Tx LS ack bytes       : 0                    Rx LS ack bytes       : 0
```

```
Total IPsec packets processed : 0
Total IPsec bytes processed   : 0
Total Number of State Changes : 0
Number of LSAs                : 0
LSA Checksum Sum              : 0

Total OSPFv3 Packets Discarded: 0
----------------------------------

Reason                           Packets Dropped
--------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
  MTU Mismatch                   0
  Neighbor Ignored               0
Resource Failures                0
Bad LSA Length                   0
Others                           0
IPsec Authentication Errors      0
IPsec ESP Errors                 0

Total LSAs Ignored : 0
-----------------------
Bad Type           : 0
Bad Length         : 0
Invalid Data       : 0
Invalid Checksum   : 0
```

summary-address

Syntax

`summary-address  <IPV6-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]`

`no summary-address <prefix/length> [no-advertise | tag <tag-value>`

Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The `no` form of this command disables route summarization.

NOTE: This command only works for an ASBR (Autonomous System Boundary Router).

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<IPV6-ADDR>

Specifies an IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

`no-advertise`

Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.

`tag` ***<TAG-VALUE>***

Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPF route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# summary-address 2001:DB8::1/32
```

Disabling route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no summary-address 2001:DB8::1/32
```

timers lsa-arrival

Syntax

`timers lsa-arrival <DELAY>`

`no timers lsa-arrival`

Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command `timers throttle lsa start` on the neighbor.

The `no` form of this command sets the LSA timers to default values.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

<DELAY>

Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no timers lsa-arrival
```

timers throttle lsa

Syntax

`timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>`

`no timers throttle lsa`

Description

Configures the timers for LSA generation.

The `no` form of this command sets the LSA timers to default values.

Command context

`config-ospfv3-<PROCESS-ID>`

Parameters

`start-time <START-TIME>`

Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.

`hold-time <HOLD-TIME>`

Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until `max-wait-time` is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

`max-wait-time <WAIT-TIME>`

Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LSA timers:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttle lsa
```

timers throttle spf

Syntax

timers throttle spf start-time *<START-TIME>* hold-time *<HOLD-TINME>*
max-wait-time *<WAIT-TIME>*

no timers throttle spf

Description

Configures timers for SPF calculation. There are three timers:

- `start-time` Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- `hold-time` Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- `max-wait-time` Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The `no` form of this command sets all the configured non-default timers to default value.

Command context

config-ospf-*<PROCESS-ID>*

Parameters

<START-TIME>

Time in milliseconds to set timer for initial SPF delay. Default: 200.

<HOLD-TINME>

Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000.

<WAIT-TIME>

Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time 3000 max-wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
        area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

transit-delay

Syntax

`transit-delay <DELAY>`

`no transit-delay`

Description

Sets the time delay in Link state transmission for virtual links.

The `no` form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

Command context

`config-router-vlink`

Parameters

<DELAY>

Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting OSPFv3 virtual links transit delay:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 virtual links transit delay to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
```

trap-enable

Syntax

`trap-enable`

`no trap-enable`

Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

The `no` form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

Command context

`config-ospfv3`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no trap-enable
```

Overview

NOTE: Key chain is not supported on the Aruba 6100 Switch Series.

This chapter provides details for configuring and verifying the functions of a key chain. Key chain configurations are spread across `config` context, `keychain` context, and `keychain-key` context.

A key chain provides seamless authentication-key rollover. When authentication method is enabled, using key chain allows OSPF to overcome static key configuration to change the key periodically. Using key chain also reduces risks of keys being guessed by configuring rotating keys.

NOTE: A key chain must be used by an application like OSPF that communicates by using the keys with its peers.

Key chain commands

accept-lifetime

Syntax

```
accept-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} | end-time <time> <month>/<day>/<year>}
```

Description

Configures the duration for which the key is valid for receiving packets.

The `no` form of this command configures the key packet receiving duration to the default value of an infinite time.

Command context

`config-keychain-key`

Parameters

`start-time`

Time at which the key chain lifetime starts. Required. Format: HH:MM:SS

`end-time`

Time at which the key chain lifetime expires. Required. Format: HH:MM:SS

`day`

Day of the month. Required. Range: 1-31.

`month`

Month of the year. Required.

year

Year. Required. Range: 2020-2050

duration

Time in seconds. Optional. Range: 1-2147483646.

infinite

Specifies infinite time for the key. Optional.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Configuring the duration for which the key is valid for receiving packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration 1000
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration infinite
switch(config-keychain-key)# accept-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime duration 1000
switch(config-keychain-key)# accept-lifetime duration infinite
```

Configuring the key packet receiving duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no accept-lifetime
```

key

Syntax

```
key <key-id>
```

Description

Creates the key for a key chain and enters the key chain key context. A maximum of 64 keys can be configured per key chain.

The `no` form of this command deletes the key from the key chain.

Command context

```
config-keychain
```

Parameters

key-id

ID of the key. Required. Range: 1-255.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Creating a key for a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
```

Deleting a key from a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# no key 1
```

keychain

Syntax

```
keychain <keychain-name>
```

Description

Creates the key chain and enters the key chain context. A maximum of 64 key chains can be configured in the system.

The `no` form of this command removes the key chain if it is not used by any subscribers.

Command context

```
config
```

Parameters

keychain-name

Name of the key chain. Required.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Creating a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Removing a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

key-string

Syntax

```
key-string {ciphertext | plaintext} <key>
```

Description

Sets the password used for the key. The password is always stored in encrypted format in the OVSDB. The key will be considered invalid if the password is not specified for the key.

The `no` form of this command deletes the password used for the key.

Command context

`config-keychain-key`

Parameters

`ciphertext`

Password in encrypted format. Required.

`plaintext`

Password in simple-text format. Required.

`key`

Password for key authentication. Required.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Setting the password for the key:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string plaintext key
  <cr>
switch(config-keychain-key)# key-string ciphertext ASDFkjdkj
  <cr>
```

Deleting the password from the key:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no key-string
  <cr>
```

send-lifetime

Syntax

```
send-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} | end-time <time> <month>/<day>/<year>}
```

Description

Configures the duration for which the key is valid for sending packets.

The `no` form of this command configures the key packet sending duration to the default value of an infinite time.

Command context

`config-keychain-key`

Parameters

start-time

Time at which the key chain lifetime starts. Required. Format: HH:MM:SS

end-time

Time at which the key chain lifetime expires. Required. Format: HH:MM:SS

day

Day of the month. Required. Range: 1-31.

month

Month of the year. Required.

year

Year. Required. Range: 2020-2050

duration

Time in seconds. Optional. Range: 1-2147483646.

infinite

Specifies infinite time for the key. Optional.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Configuring the duration for which the key is valid for sending packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration 1000
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration infinite
switch(config-keychain-key)# send-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime duration 1000
switch(config-keychain-key)# send-lifetime duration infinite
```

Configuring the key packet sending duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no send-lifetime
```

show capacities keychain

Syntax

```
show capacities keychain
```

Description

Shows the maximum number of key chains and keys configurable in a key chain.

Command context

Operator (>)

Parameters

None

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show capacities keychain

System Capacities: Filter Keychain
Capacities Name                                                                                      Value
-----------------------------------------------------------------------------------------------------------
Maximum number of keychains supported in the system                                                   64
Maximum number of Keys supported in a single Keychain                                                 64
```

show keychain

Syntax

```
show keychain [<keychain-name>]
```

Description

Shows information about configured and active keys of a named key chain or (if `keychain-name` is not specified) all configured key chains.

Command context

Operator (>)

Parameters

keychain-name

Name of the key chain. Optional.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show keychain
Keychain Name : ospf_keys
  Number of Keys       : 2
  Active Send Key ID   : 7
  Active Recv Key IDs : 7, 200

  Key ID               : 7
    Key string         : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to infinite
  Key ID               : 200
    Key string         : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

Keychain Name : bgp_keys
  Number of Keys       : 2
  Active Send Key ID   : 7
```

```
  Active Recv Key IDs : 7

  Key ID              : 7
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/26/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/22/2020 to infinite
  Key ID              : 8
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021


Keychain Name : ospf_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7, 200

  Key ID              : 7
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to infinite
  Key ID              : 200
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
```

show running-config keychain

Syntax

```
show runnning-config keychain
```

Description

Shows the configurations for key chain protocol.

Command context

Operator (>)

Parameters

None

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show running-config keychain
keychain ospf_keys
  key 1
    key-string ciphertext AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
    send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 45
    key-string ciphertext AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
```

```
  key 33
switch#
```

Overview

The client IP address tracking feature will learn and update the IP addresses of the access devices and clients connected to the switch. It can track addresses of directly connected clients, as well as clients connected to a downstream device such as a wireless access point.

IP Client Tracker commands

client track ip

Syntax

```
client track ip
```

Description

Enables client IP address tracking on the switch. The default is disabled on global and VLAN levels.

Admin users can enable client IP address tracking at the VLAN level.

NOTE: Tracking enabling will take effect only if the client IP address tracking is enabled at system and VLAN level.

The `no` form of the command disables client IP address tracking. If tracking is disabled at switch level, it will be stopped even if it is enabled at VLAN or port level.

Command context

```
config
```

Parameters

None

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enable client IP address tracking at switch level:

```
switch(config)# client track ip
```

Enable client IP address tracking on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# client track ip
Enable client IP address tracking on VLANs 10 to 100:
switch(config)# vlan 10-100
switch(config-vlan-<10-100>)# client track ip
```

client track ip { enable | disable | auto }

Syntax

```
client track ip { enable | disable | auto }
```

Description

Enables client IP address tracking on the specified set of interfaces. Tracking will take effect only if client IP address tracking is enabled at both the system level and for the VLAN to which the port belongs. Default: auto.

The `no` form of the command disables client IP address tracking on the specified set of interfaces.

Command context

`config-if`

Parameters

enable

Specifies that all client IP addresses will be tracked in the port.

disable

Specifies that client IP addresses will not be tracked in the port.

auto

Specifies the following:

- For LLDP devices: Only the specified client IP address will be tracked in the port and other client IP addresses will not be tracked.
- For non-LLDP devices: All client IP addresses will be tracked in the port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enable client IP address tracking on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip enable
```

Enable client IP address tracking on interfaces 1/1/1 to 1/1/5:

```
switch(config)# interface 1/1/1-1/1/5
switch(config-if-<1/1/1-1/1/5>)# client track ip enable
```

client track ip client-limit

Syntax

```
client track ip client-limit <CLIENT-LIMIT>
```

Description

Configures the maximum number of clients to be tracked on the specified set of interfaces.

For the 6100 Switch Series the `no` form of the command resets the client limit to the default value of 128.

For the 6200 Switch Series the `no` form of the command resets the client limit to the default value of 2048.

Command context

```
config-if
```

Parameters

CLIENT-LIMIT

Specifies the maximum number of clients tracked on a port. Required. For the 6100 Switch Series, Range: 1-128. Default: 128.For the 6200 Switch Series, Range: 1-2048. Default: 2048..

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configure the maximum number of clients to be tracked on interface 1/1/5:

```
switch(config)# interface 1/1/5
switch(config-if)# client track ip client-limit 32
```

client track ip update-interval

Syntax

```
client track ip update-interval <INTERVAL>
```

Description

Configures how often client IP addresses are updated.

The `no` form of the command resets the update interval to the default of 1800 seconds.

Command context

```
config-if
```

Parameters

INTERVAL

Specifies the update interval in seconds. Required. Range: 60-28000. Default: 1800.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configure the update interval for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip update-interval 600
```

client track ip update-method probe

Syntax

```
client track ip update-method probe
```

Description

Enables probing the client to update the IP address.

The probe is sent to all clients on the tracking list that have an IP address in the following scenarios:

1. IP packets are not received from the clients during the IP address update cycle.
2. There is no IP packet from a learned IP address. In this case, a probe will be sent for the IP address to confirm if it is still owned by that client.

The `no` form of the command disables probing.

Command context

`config`

Parameters

None

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Disable probing to update the client IP address:

```
switch(config)# no client track ip update-method probe
```

show capacities

Syntax

```
show capacities
```

Description

Shows the capacities configured on the switch.

Command context

Operator (>)

Parameters

None

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show capacities

System Capacities:
Capacities Name                                                                                      Value
-------------------------------------------------------------------------------------------------------------
Maximum number of Access Control Entries configurable in a system                                   14336
Maximum number of Access Control Lists configurable in a system                                      1024
Maximum number of class entries configurable in a system                                             1024
Maximum number of classes configurable in a system                                                    512
```

```
Maximum number of entries in an Access Control List                                                    1024
Maximum number of entries in a class                                                                   1024
Maximum number of entries in a policy                                                                  1024
Maximum number of classifier policies configurable in a system                                          512
Maximum number of policy entries configurable in a system                                              1024
Maximum number of clients supported for tracking the IP address in the system                           128
switch# show capacities client-track-ip-client-limit

System Capacities: Filter Client Track IP Client Limit
Capacities Name                                                                                       Value
-----------------------------------------------------------------------------------------------------------
Maximum number of clients supported for tracking the IP address in the system
```

show client ip { count | port | vlan }

Syntax

```
show client ip { count | port | vlan }
```

Description

Shows number of client IP addresses or information about client IP addresses tracked on ports and VLANs.

Command context

Operator (>)

Parameters

count

Displays number of clients tracked.

port

Displays client IP addresses tracked on the ports.

vlan

Displays client IP addresses tracked on the VLANs.

Authority

Administrators or local user group members with execution rights for this command.

Overview

NOTE: Routing Information Protocol (RIP, RIPv2 and RIPng) are not supported on the Aruba 6100 Switch Series.

Routing Information protocol (RIP, RIPv2, RIPng) is a distance-vector routing protocol that uses hop count as a routing metric. It is useful in small networks. RIP can be configured on L3 ports, LAG, VLAN and Loopback interfaces. All configurations work in the interface context and require the associated interface to be routing interface.

- **RIP/RIPv2:** IPv4 implementation of Routing Information protocol defined in RFC 2453.
- **RIPng:** IPv6 implementation of Routing Information protocol defined in RFC 2080.

RIP characteristics:

- Prevents routing loops by adding a limit to the number of hops allowed in a path from source to destination.
- Allows a maximum of 15 hops which limits the size of the network that RIP can support.
- Uses broadcast UDP data packets to exchange routing information.
- RIPv2 is a classless protocol that supports variable-length subnet mask (VLSM), classless inter-domain routing (CIDR) and route summarization.
- RIPng uses ipsec for message authentication.

RIP limitations:

- RIPv1 is not supported.
- Only poison reverse is supported for loop prevention.

RIPv2 (IPv4) commands

Configuration commands

`router rip`

Syntax

```
router rip <PROCESS-ID> [vrf <VRF-NAME>]

no router rip <PROCESS-ID> [vrf <VRF-NAME>]
```

Description

Creates RIP process if not already created and enters the `router rip <PROCESS-ID>` context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIP process is allowed per VRF.

The `no` form of this command deletes the RIP instance for the VRF. If no VRF is mentioned the default is deleted.

Command context

`config`

Parameters

<PROCESS-ID>

Specifies name of the RIP process ID. Range: 1-63.

vrf <VRF-NAME>

Specifies VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating RIP process and naming the VRF:

```
switch(config)# router rip 2 vrf red
```

Deleting RIP process:

```
switch(config)# no router rip 2 vrf red
```

Interface commands

ip rip

Syntax

`ip rip <PROCESS-ID> {all-ip | ip-address}`

`no ip rip <PROCESS-ID> {all-ip | ip-address}`

Description

Enables RIP process on an interface.

The `no` form of this command deletes the RIP process from an interface.

Command context

`config-if`

Parameters

ip rip *<PROCESS-ID>*

Specifies RIP process ID. Range: 1-63.

all-ip

Specifies RIP for all IP addresses configured on the interface.

ip-address

Specifies IP address for RIP on the interface.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.
- If `ip rip 1 all-ip` is configured and a new IP address is added to the interface, RIP configurations will not be applicable for the newly added IP address.

Examples

Configuring RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 all-ip
```

Deleting RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip 1 all-ip
```

ip rip all-ip enable

Syntax

`ip rip all-ip enable`

`no ip rip all-ip enable`

Description

Enables RIP process for all RIP enabled IP addresses configured on interface.

The `no` form of this command disables RIP process on the interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

ip rip all-ip disable

Syntax

```
ip rip all-ip disable

no ip rip all-ip disable
```

Description

Disables RIP process for all RIP enabled IP addresses configured on the interface.

The `no` form of this command enables RIP process for all RIP enabled IP addresses configured on the interface.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

ip rip all-ip send disable

Syntax

```
ip rip all-ip send disable

no ip rip all-ip send disable
```

Description

Disables interface from sending RIP packets for all RIP enabled IP addresses.

The `no` form of this command enables interface to send RIP packets for all RIP enabled IP addresses.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to send RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from sending RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip send disable
```

Enabling interface to send RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip send disable
```

ip rip all-ip receive disable

Syntax

```
ip rip all-ip receive disable

no ip rip all-ip receive disable
```

Description

Disables interface from receiving RIP packets for all enabled IP addresses.

The `no` form of this command enables interface to receive RIP packets for all RIP enabled IP addresses.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from receiving RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip receive disable
```

Enabling interface to receive RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip receive disable
```

Routing commands

enable

Syntax

`enable`

`no enable`

Description

Enables RIP process if disabled. By default RIP process is enabled.

The `no` form of this command disables the RIP process.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling RIP process when disabled:

```
switch(config)# router rip 1
switch(config-rip-1)# enable
```

Disabling RIP process when enabled:

```
switch(config)# router rip 1
switch(config-rip-1)# no enable
```

disable

Syntax

`disable`

`no disable`

Description

Disables RIP process.

The `no` form of this command enables the RIP process.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# disable
```

Enabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# no disable
```

distance

Syntax

`distance <DISTANCE>`

`no distance`

Description

Configures administrative distance for RIP. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The `no` form of this command sets the RIP administrative distance to the default. Default: 120.

Command context

`config`

Parameters

<DISTANCE>

Specifies RIP administrative distance. Range: 1 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring administrative distance for RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# distance 100
```

Setting administrative distance for RIP to default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no distance
```

maximum-paths

Syntax

`maximum-paths <MAX-VALUE>`

`no maximum-paths`

Description

Sets the maximum number of ECMP routes that RIP can support.

The `no` form of this command sets the maximum number of ECMP routes to the default value of 4.

Command context

`config`

Parameters

<MAX-VALUE>

Sets the number of RIP ECMP routes. Range: 1-8.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting maximum number of RIP ECMP routes:

```
switch(config)# router rip 1
switch (config-rip-1)# maximum-paths 8
```

Setting maximum number of RIP ECMP routes to default:

```
switch(config)# router rip 1
switch (config-rip-1)# no maximum-paths
```

redistribute

Syntax

`redistribute {bgp | connected | ospf <PROCESS-ID> | static}`

`no redistribute {bgp | connected | ospf <PROCESS-ID> | static}`

Description

Redistributes routes originating from other protocols into RIP.

The `no` form of this command disables redistribution of routes originating from other protocols into RIP.

Command context

`config`

Parameters

bgp

Specifies BGP routes to redistribute into RIP.

connected

Specifies connected routes (directly attached subnet or host) to redistribute into RIP.

ospf *<PROCESS-ID>*

Specifies OSPF route to redistribute into RIP.

static

Specifies static route to redistribute into RIP.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Redistributing BGP routes into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# no redistribute bgp
```

timers update

Syntax

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>

no timers
```

Description

Configures RIP timers with specific values.

The `no` form of this command sets all RIP timers to default values.

Command context

`config`

Parameters

timers update *<INTERVAL>*

Specifies frequency at which RIP sends updates to all of its peers. Range: 1 to 2147484. Default: 30.

timeout *<DURATION>*

Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180.

garbage-collection *<PERIOD>*

Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RIP timers with specific values:

```
switch(config)# router rip 1
switch(config-rip-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIP timers with default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no timers
```

RIPv2 clear commands

clear ip rip statistics

Syntax

`clear ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

Description

Clears RIP event statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63

all-vrfs

Clears statistics for all VRFs.

vrf

Selects VRF to clear statistics for.

<VRF-NAME>

Specifies VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing RIP event statistics:

```
switch# clear ip rip statistics
```

RIPv2 interface commands

enable

Syntax

`enable`

`no enable`

Description

Enables RIP process for RIP enabled IP address configured on interface.

The `no` form of this command disables RIP process on interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Enabling RIP process for RIP enabled IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no enable
```

disable

Syntax

`disable`

`no disable`

Description

Disables RIP process for RIP enabled IP addresses configured on interface.

The `no` form of this command enables RIP process on interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling RIP process for RIP enabled IP addresses configured on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# disable
```

Enabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no disable
```

send disable

Syntax

```
send disable

no send disable
```

Description

Disables an interface from sending RIP packets for a specific IP address.

The `no` form of this command enables interface for sending RIP packets for a specific IP address.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to send and receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from sending RIP packets for a specific IP address :

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# send disable
```

Enabling interface to send RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no send disable
```

receive disable

Syntax

```
receive disable

no receive disable
```

Description

Disables interface from receiving RIP packets for a specific IP address.

The no form of this command enables interface for receiving RIP packets for a specific IP address.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# receive disable
```

Enabling interface for receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no receive disable
```

RIPv2 show commands

show capacities rip

Syntax

```
show capacities rip
```

Description

Displays maximum number of RIP interfaces, routes and process.

Command context

Operator (>) or Manager (#)

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Displaying maximum number of RIP interfaces, routes and process:

```
switch# show capacities rip

System Capacities: Filter RIP
Capacities Name                                                                                 Value
-----------------------------------------------------------------------------------------------------
Maximum number of RIP interfaces configurable in the system                                        32
Maximum number of RIP processes supported across each VRF                                           1
Maximum number of routes in RIP supported across all VRFs                                        2540
```

show capacities-status rip

Syntax

`show capacities-status rip`

Description

Displays number of RIP interfaces, routes and process configured in the system.

Command context

Operator (>) or Manager (#)

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Displaying number of RIP interfaces, routes and process:

```
switch# show capacities-status rip

System Capacities Status: Filter RIP
Capacities Name                                                                         Value Maximum
-----------------------------------------------------------------------------------------------------
Number of RIP interfaces configured in the system                                           0      32
```

show ip rip

Syntax

`show ip rip [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays general RIP configuration.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63.

all vrfs

Displays general RIP information for all VRFs.

vrf

Selects VRF to display general RIP information for.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters display general RIP information for a specific RIP process.
- Parameters display general RIP information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying general RIP configuration for all VRFs:

```
switch# show ip rip 34 all-vrfs
VRF : Default                      Process-ID : 34
----------------------------------------------------------------------
RIP Version             : RIPv2     Protocol Status : Enabled
Update Time             : 60  sec   Timeout Time    : 240 sec
Garbage Collection Time : 250 sec   ECMP            : 6
Distance                : 100       Redistribution  : static,
                                                      ospf 1

VRF : vrf_1                        Process-ID : 34
----------------------------------------------------------------------
RIP Version             : RIPv2     Protocol Status : Enabled
Update Time             : 30  sec   Timeout Time    : 180 sec
Garbage Collection Time : 120 sec   ECMP            : 4
Distance                : 120       Redistribution  : None
```

show ip rip interface

Syntax

`show ip rip [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays information about RIP enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63.

<INTERFACE-NAME>

Specifies interface.

brief

Shows brief overview information for the RIP interface.

all-vrfs

Displays interface information for all VRFs.

vrf

Selects specific VRF.

<VRF-NAME>

Specifies VRF.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters display general RIP information for a specific RIP process.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

```
switch# show ip rip interface
Interface 1/1/1 is up, IP Address is 10.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default            Process-ID     : 1
Status          : Oper Up            Mode           : Send and Receive
MTU             : 500                Version        : RIPv2
Poision Reverse : Enabled

Interface 1/1/2 is up, IP Address is 20.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default            Process-ID     : 1
Status          : Admin Down         Mode           : Receive
MTU             : 500                Version        : RIPv2
Poision Reverse : Enabled

Interface 1/1/3 is up, IP Address is 30.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default            Process-ID     : 1
Status          : Admin Down         Mode           : Send
MTU             : 500                Version        : RIPv2
Poision Reverse : Enabled

switch# show ip rip interface brief
VRF : default   Process-ID : 1
-------------------------------

Total Number of Interfaces: 2
```

```
Interface      IP-Address/Mask    Status    MTU
--------------------------------------------------
1/1/1          10.10.10.1/24       up        500
1/1/2          20.10.10.1/24       up        500
1/1/3          30.10.10.1/24       up        500
```

show ip rip neighbors

Syntax

`show ip rip [<PROCESS-ID>] neighbors [<IP-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays information about RIP neighbors.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63.

<IP-ADDRESS>

Specifies IP address of a specific neighbor to display information on.

all-vrfs

Displays neighbor information for all VRFs.

vrf

Selects VRF to display neighbor information.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters display RIP neighbor information for a specific RIP process.
- Parameters display RIP neighbor information for a specific neighbor.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP neighbor information for all VRFs:

```
switch# show ip rip neighbors all-vrfs
VRF : default            Process-ID : 1
----------------------------------------

Total Number of Neighbors: 1

Peer-Address    Type    Last-Update  Rcvd-Bad-Pkts    Rcvd-Bad-Routes
```

```
-----------------------------------------------------------------
1.1.1.2        RIPv2    0:0:7        4                5
```

show ip rip routes

Syntax

show ip rip [*<PROCESS-ID>*] routes [*<PREFIX/LENGTH>*] [all-vrfs | vrf *<VRF-NAME>*]

Description

Displays RIP routing table for a specific RIP process.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID to display information for a specific RIP process. Range: 1-63.

<PREFIX/LENGTH>

Specifies the network prefix.

all-vrfs

Displays RIP routing information for all VRFs.

vrf

Selects VRF to display RIP routing information.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- *<PREFIX/LENGTH>* is an optional parameter that displays RIP routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP routing table for all VRFs:

```
switch# show ip rip routes all-vrfs
VRF : default              Process-ID : 1
----------------------------------------
Total Number of Routes : 6

Prefix              Metric     Interface      Nexthop
--------------------------------------------------------
10.1.0.0/16           2          1/1/1          30.1.1.2
20.1.2.0/24           3          1/1/1          30.1.1.2
30.1.1.0/24           1          1/1/1
```

```
VRF : vrf_1                Process-ID : 34
---------------------------------------

Prefix                  Metric     Interface      Nexthop
-------------------------------------------------------
20.1.0.0/16               10         1/1/2          50.1.1.2
40.1.2.0/24               14         1/1/2          50.1.1.2
50.1.1.0/24                1         1/1/2
```

show ip rip statistics

Syntax

`show ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

Description

Displays RIP statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63.

all-vrfs

Displays statistics information for all VRFs.

vrf

Selects VRF to display RIP statistics information for.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP statistics for all VRFs:

```
switch# show ip rip statistics all-vrfs
VRF : default  Process-ID : 1
-------------------------------------

Global Route Changes : 50
Global Queries       : 2
Last Cleared         : 0h 30m 28s ago

VRF : vrf_1    Process-ID : 34
```

```
-------------------------------------

Global Route Changes : 20
Global Queries       : 0
Last Cleared         : 0h 30m 28s ago
```

show ip rip statistics interface

Syntax

`show ip rip [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays RIP statistics for RIP enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63.

<INTERFACE-NAME>

Specifies name of interface.

all-vrfs

Displays RIP interface statistics for all VRFs.

vrf

Selects VRF to display RIP interface statistics.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP statistics for a RIP enabled interface:

```
switch# show ip rip statistics interface 1/1/1
VRF : default  Process-ID : 1      interface 1/1/1
--------------------------------------------------

IP-Address    Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
-----------------------------------------------------------------------
10.1.1.1      15                 3                   4
```

```
Last Cleared : 0h 30m 28s ago
```

show running-config

Syntax

```
show running config
```

Description

Displays all running configurations for all protocols including RIP.

Command context

Operator (>) or Manager (#)

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Displaying all running configurations for all protocols including RIP:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf blue
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf red
router rip 1
    maximum-paths 5
    distance 1
router rip 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
    redistribute ospf 1
    timers update 40 timeout 200 garbage-collection 120
vlan 1
```

```
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip rip 1 33.1.1.1
        send disable
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip address 44.44.1.1/24 secondary
    ip rip 1 33.44.1.1
        send disable
    ip rip 1 44.44.1.1
        send disable
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip rip 1 55.55.55.55
```

RIPng (IPv6) commands

Configuration commands

`router ripng`

Syntax

`router ripng <PROCESS-ID> [vrf <VRF-NAME>]`

`no router ripng <PROCESS-ID> [vrf <VRF-NAME>]`

Description

Creates RIPng process if not already created and enters the `router ripng <PROCESS-ID>` context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIPng process is allowed per VRF.

The `no` form of this command deletes the RIPng instance for the VRF. If no VRF is mentioned the default is deleted.

Command context

`config`

Parameters

<PROCESS-ID>

Specifies name of the RIPng process ID. Range: 1-63.

vrf

Sets VRF name for RIPng process.

<VRF-NAME>

VRF name for VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating RIPng process and naming the VRF:

```
switch(config)# router ripng 2 vrf red
```

Deleting RIPng process:

```
switch(config)# no router ripng 2 vrf red
```

Interface commands

ipv6 ripng

Syntax

```
ipv6 ripng <PROCESS-ID>

no ipv6 ripng <PROCESS-ID>
```

Description

Enables RIPng process on interface and creates a new context.

The `no` form of this command deletes RIPng process on interface.

Command context

`config-if`

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)#
```

Deleting RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# no ipv6 ripng 1
```

Routing commands

enable

Syntax

```
enable

no enable
```

Description

Enables RIPng process if disabled. By default RIPng process is enabled.

The `no` form of this command disables the RIPng process.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling RIPng process when disabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# enable
```

Disabling RIPng process when enabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no enable
```

disable

Syntax

```
disable

no disable
```

Description

Disables RIPng process.

The `no` form of this command enables the RIPng process.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# disable
```

Enabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no disable
```

distance

Syntax

```
distance <DISTANCE>

no distance
```

Description

Configures administrative distance for RIPng. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The `no` form of this command sets the RIPng administrative distance to the default. Default: 120.

Command context

`config`

Parameters

<DISTANCE>

Specifies RIPng administrative distance. Range: 1 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring administrative distance for RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# distance 100
```

Setting administrative distance for RIPng to default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no distance
```

maximum-paths

Syntax

```
maximum-paths <MAX-VALUE>

no maximum-paths
```

Description

Sets the maximum number of ECMP routes that RIPng can support.

The `no` form of this command sets the maximum number of ECMP routes to the default value of 4.

Command context

`config`

Parameters

<MAX-VALUE>

Sets the number of RIPng ECMP routes. Range: 1-8.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting maximum number of RIPng ECMP routes:

```
switch(config)# router ripng 1
switch (config-ripng-1)# maximum-paths 8
```

Setting maximum number of RIPng ECMP routes to default:

```
switch(config)# router ripng 1
switch (config-ripng-1)# no maximum-paths
```

redistribute

Syntax

`redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}`

`no redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}`

Description

Redistributes routes originating from other protocols into RIPng.

The `no` form of this command disables redistribution of routes originating from other protocols into RIPng.

Command context

`config`

Parameters

`bgp`

Specifies BGP routes to redistribute into RIPng.

`connected`

Specifies connected routes (directly attached subnet or host) to redistribute into RIPng.

`ospfv3 <PROCESS-ID>`

Specifies OSPFv3 route to redistribute into RIPng.

`static`

Specifies static route to redistribute into RIPng.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Redistributing BGP routes into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no redistribute bgp
```

timers update

Syntax

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>

no timers
```

Description

Configures RIPng timers with specific values.

The `no` form of this command sets all RIPng timers to default values.

Command context

`config`

Parameters

timers update *<INTERVAL>*

Specifies frequency at which RIPng sends updates to all of its peers. Range: 1 to 2147484. Default: 30.

timeout *<DURATION>*

Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180.

garbage-collection *<PERIOD>*

Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RIPng timers with specific values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIPng timers with default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no timers
```

RIPng clear commands

clear ipv6 ripng statistics

Syntax

```
clear ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

Description

Clears RIPng event statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63

all-vrfs

Clears statistics for all VRFs.

vrf

Selects VRF to clear statistics for.

<VRF-NAME>

Specifies VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing RIPng event statistics:

```
switch# clear ipv6 ripng statistics
```

RIPng interface commands

enable

Syntax

`enable`

`no enable`

Description

Enables RIPng process on interface.

The `no` form of this command disables RIPng process on interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# enable
```

Disabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no enable
```

disable

Syntax

`disable`

`no disable`

Description

Disables RIPng process on interface.

The `no` form of this command enables RIPng process on interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# disable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no disable
```

send disable

Syntax

`send disable`

`no send disable`

Description

Disables interface from sending RIPng packets. An interface can send RIPng packets by default.

The `no` form of this command enables interface to send RIPng packets, if disabled.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling interface from sending RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# send disable
```

Enabling interface to send RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# no send disable
```

`receive disable`

Syntax

```
receive disable

no receive disable
```

Description

Disables interface from receiving RIPng packets for all enabled IP addresses. An interface can receive RIPng packets by default.

The `no` form of this command enables interface to receive RIPng packets, if disabled.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling interface from receiving RIPng packets:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# receive disable
```

Enabling interface to receive RIPng packets when disabled:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# no receive disable
```

RIPng show commands

`show capacities ripng`

Syntax

```
show capacities ripng
```

Description

Displays the maximum number of RIPng interfaces, routes and process.

Command context

Operator (>) or Manager (#)

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Displaying maximum number of RIPng interfaces, routes and process:

```
switch# show capacities ripng

System Capacities: Filter RIPng
Capacities Name                                                                                     Value
---------------------------------------------------------------------------------------------------------
Maximum number of RIPng interfaces configurable in the system                                          32
Maximum number of RIPng processes supported across each VRF                                             1
Maximum number of routes in RIPng supported across all VRFs                                          2540
```

show capacities-status ripng

Syntax

`show capacities-status ripng`

Description

Displays number of RIPng interfaces, routes and process configured in the system.

Command context

Operator (>) or Manager (#)

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Displaying number of RIPng interfaces, routes and process:

```
switch# show capacities-status ripng

System Capacities Status: Filter RIPng
Capacities Name                                                                     Value  Maximum
--------------------------------------------------------------------------------------------------
Number of RIPng interfaces configured in the system                                     0       32
```

show ipv6 ripng

Syntax

`show ipv6 ripng [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays general RIPng configuration.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

all vrfs

Displays general RIPng information for all VRFs.

vrf

Selects VRF to display general RIPng information for.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying general RIPng configuration for all VRFs:

```
switch# show ipv6 ripng 34 all-vrfs
VRF : Default                      Process-ID : 34
---------------------------------------------------------------------
Protocol Status          : Enabled   ECMP            : 6
Update Time              : 60  sec   Timeout Time    : 240 sec
Garbage Collection Time : 250 sec   Distance        : 100
Redistribution           : static,
                           ospfv3 1

VRF : vrf_1                        Process-ID : 34
---------------------------------------------------------------------
Protocol Status          : Enabled   ECMP            : 4
Update Time              : 30  sec   Timeout Time    : 180 sec
Garbage Collection Time : 120 sec   Distance        : 120
Redistribution           : None
```

show ipv6 ripng interface

Syntax

```
show ipv6 ripng [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays information about RIPng enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

<INTERFACE-NAME>

Specifies interface.

brief

Shows brief overview information for RIPng interface.

all-vrfs

Displays interface information for all VRFs.

vrf

Selects specific VRF.

<VRF-NAME>

Specifies VRF.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

```
switch# show ipv6 ripng interface
Interface 1/1/1 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Oper Up            Mode            : Send and Receive
MTU              : 500                Poision Reverse : Enabled

Interface 1/1/2 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Admin Down         Mode            : Receive
MTU              : 500                Poision Reverse : Enabled

Interface 1/1/3 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Admin Down         Mode            : Send
MTU              : 500                Poision Reverse : Enabled

switch# show ipv6 ripng interface brief
VRF : default   Process-ID : 1
-------------------------------

Total Number of Interfaces: 2

Interface     IPv6-Address                      Status    MTU
```

```
----------------------------------------------------------
1/1/1          fe80::7272:cfff:fe70:67a       up         500
1/1/2          fe80::7272:cfff:fe71:67a       up         500
```

show ipv6 ripng neighbors

Syntax

`show ipv6 ripng [<PROCESS-ID>] neighbors [<LINK-LOCAL-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays information about RIPng neighbors.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

neighbors

Specifies neighbor IP address.

<LINK-LOCAL-ADDRESS>

Specifies link-local address.

all-vrfs

Displays neighbor information for all VRFs.

vrf

Selects VRF to display neighbor information.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters display RIPng neighbor information for a specific RIPng process.
- Parameters display RIPng neighbor information for a specific neighbor.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng neighbor information for all VRFs:

```
switch# show ipv6 ripng neighbors all-vrfs
VRF : default              Process-ID : 1
----------------------------------------
```

```
Total Number of Neighbors: 1

Peer-Address      Type    Last-Update  Rcvd-Bad-Pkts    Rcvd-Bad-Routes
----------------------------------------------------------------------
fe80::7272:cfff:fe70:86ae
                  RIPng   0:0:7        4                5
```

show ipv6 ripng routes

Syntax

`show ipv6 ripng [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays RIPng routing table for a specific RIPng process.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID to display information for a specific RIPng process. Range: 1-63.

<PREFIX/LENGTH>

Specifies the network prefix.

all-vrfs

Displays RIPng routing information for all VRFs.

vrf

Selects VRF to display RIPng routing information.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- *<PREFIX/LENGTH>* is an optional parameter that displays RIPng routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng routing table for all VRFs:

```
switch# show ipv6 ripng routes all-vrfs
VRF : default              Process-ID : 1
------------------------------------------

Prefix              Metric     Interface      Nexthop
-----------------------------------------------------------------------
2001:DB8:10::/64      2          1/1/1          FE80::2E0:E6FF:FE1B:8242
```

```
2002:DB8:10::/64      3         1/1/1         FE80::2E0:E6FF:FE1B:8242
2003:DB8:10::/64      1         1/1/1

VRF : vrf_1                Process-ID : 34
-------------------------------------------

Prefix                 Metric     Interface      Nexthop
------------------------------------------------------------------------
3001:DB8:10::/64      10         1/1/2         FE80::2E0:E6FF:FE1B:8232
3002:DB8:10::/64      14         1/1/2         FE80::2E0:E6FF:FE1B:8232
3003:DB8:10::/64       1         1/1/2
```

show ipv6 ripng statistics

Syntax

`show ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

Description

Displays RIPng statistics.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

all-vrfs

Displays statistics information for all VRFs.

vrf

Selects VRF and displays RIPng statistics for it.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng statistics for all VRFs:

```
switch# show ipv6 ripng statistics all-vrfs
VRF : default  Process-ID : 1
-------------------------------------

Global Route Changes : 50
Global Queries       : 2
```

```
Last Cleared          : 0h 30m 28s ago

VRF : vrf_1    Process-ID : 34
-------------------------------------

Global Route Changes : 20
Global Queries       : 0
Last Cleared         : 0h 30m 28s ago
```

show ipv6 ripng statistics interface

Syntax

`show ipv6 ripng [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]`

Description

Displays RIPng statistics for RIPng enabled interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

<INTERFACE-NAME>

Specifies name of interface.

all-vrfs

Displays RIPng interface statistics for all VRFs.

vrf

Selects VRF to display RIPng interface statistics.

<VRF-NAME>

Specifies VRF name.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng statistics for a RIPng enabled interface:

```
switch# show ipv6 ripng statistics interface 1/1/1
VRF : default  Process-ID : 1      interface 1/1/1
--------------------------------------------------

IPv6-Address         Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
-----------------------------------------------------------------------------
```

```
fe80::7272:cfff:fe70:86ae
                     15                  3                    4

Last Cleared: 0h 30m 28s ago
```

show running-config

Syntax

show running config

Description

Displays all running configurations for all protocols including RIPng.

Command context

Operator (>) or Manager (#)

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Displaying all running configurations for all protocols including RIPng:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospfv3 1
router ripng 1
    maximum-paths 5
    distance 1
    redistribute ospfv3 1
    timers update 40 timeout 200 garbage-collection 150
router ripng 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
```

```
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        send disable
interface 1
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        receive disable
```

Accessing Aruba Support

Aruba Support Services	**https://www.arubanetworks.com/support-services/**
Aruba Support Portal	**https://asp.arubanetworks.com/**
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	**https://www.arubanetworks.com/support-services/contact-support/**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
Software licensing	**https://lms.arubanetworks.com/**
End-of-Life information	**https://www.arubanetworks.com/support-services/end-of-life/**
Aruba software and documentation	**https://asp.arubanetworks.com/downloads**

Accessing updates

To download product updates:

Aruba Support Portal

https://asp.arubanetworks.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

IMPORTANT: Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://asp.arubanetworks.com/notifications/subscriptions (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

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